Supplementary Technical Material: Relativistic Quantum Information Review

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This supplement retains detailed derivations, numerical reconstructions, and technical audits that support the concise main review. Cross-references to equations and sections in the main manuscript are resolved through the accompanying source file.

S1 Matched benchmark for comparing measurement frameworks

The approaches discussed above should be compared on a common operational task, rather than by placing their formal expressions side by side. We specify a benchmark that can be implemented first for a free real scalar field in four-dimensional Minkowski spacetime and then extended to curved backgrounds or gauge fields. The benchmark is deliberately modest: it does not assert that QTP, detector instruments, the positive formalism, and local S-matrix constructions are equivalent. It identifies which predictions can be matched and which additional structures each framework supplies.

Let $\hat{\phi}(x)$ be a free scalar field in the Minkowski vacuum $|0\rangle$, and let two initially uncorrelated two-level probes A and B have gaps Ω_A and Ω_B. Their interaction regions are compact spacetime sets $\mathcal{O}_A$ and $\mathcal{O}_B$. For a finite-size probe, introduce real smooth test functions $F_\nu(x)$ supported in $\mathcal{O}_\nu$, with $\nu = A, B$. A convenient interaction-picture coupling is

$$H_I(t) = \lambda \sum_{\nu=A,B} \chi_\nu(t)\, \hat{\mu}_\nu(t)\, \hat{\Phi}_\nu(t), \qquad \hat{\Phi}_\nu(t) = \int d^3\mathbf{x}\, f_\nu(t,\mathbf{x}) \hat{\phi}(t,\mathbf{x}), \tag{S1}$$

where $\hat{\mu}_\nu(t) = \hat{\sigma}_\nu^+ e^{i\Omega_\nu t} + \hat{\sigma}_\nu^- e^{-i\Omega_\nu t}$, and the spacetime support of $\chi_\nu f_\nu$ is contained in $\mathcal{O}_\nu$. Equivalently, the coupling can be written using a single spacetime test function F_ν. The field input, probe preparation, smearing, switching, coupling strength, and readout are held fixed when changing the measurement framework.

For one probe initially in its ground state, the leading excitation probability is

$$P_\nu^{(2)} = \lambda^2 \int d^4x\, d^4x'\, F_\nu(x) F_\nu(x')\, e^{-i\Omega_\nu[t(x)-t(x')]} W_0(x,x'), \tag{S2}$$

up to the convention used to absorb the switching and spatial profile into F_ν. Here $W_0(x,x') = \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle$ is the Wightman function. Equation (S2) is the common reference quantity for the first comparison; a detector model may add a finite readout window, internal noise, or a nontrivial response function.

In QTP, the corresponding primary output is a time-resolved event probability density. The detector response kernel weights the same unequal-time field correlator, and an appropriate integration over the recorded detection time and detector response should recover the total transition probability in the regime where the QTP model reduces to the chosen detector coupling [1, 2]. This is a matching statement about a specified response model, not an identity between all QTP prescriptions and all Unruh–DeWitt detectors. The advantage of QTP is

that it retains the detection-time variable and generalizes naturally to joint event probabilities involving higher-point unequal-time correlators.

The positive-formalism construction addresses a different part of the benchmark. It supplies a probabilistically consistent and compositional description of localized measurements, including a specified class of quadratic observables, and makes locality and composition explicit [3, 4]. In the Oeckl–Zampeli construction, however, a modulus-square probe for several observables encodes a joint expectation rather than a joint distribution of simultaneous outcomes; probabilities are obtained separately from the positive formalism's ratio rule. To compare its output with Eq. (S2), one must specify a probe and outcome functional corresponding to the same smeared observable and readout. Agreement of a single-measurement probability is a necessary check, but the distinctive test is whether two localized measurement probes compose consistently when their spacetime regions are combined or separated. The positive formalism does not, by itself, select the same two-level detector state or define its negativity.

The local S-matrix approach supplies yet another test. The same smeared coupling is used to construct local operations and, where an explicit pointer is included, field-induced Kraus operators. The question is whether the operations satisfy the relevant causal factorization conditions and whether the resulting conditional updates avoid signalling in configurations involving multiple regions [5]. This is not a second formula for the detector excitation probability: it is a test of the causal admissibility and compositional behavior of the operation that produces the record. In a time-extended pointer model, the retarded self-propagator also constrains the attainable measurement resolution, so the pointer width and readout noise must be included in a fair comparison.

The second stage uses both probes. We require the full supports $\mathcal{O}_A$ and $\mathcal{O}_B$ to be strictly spacelike separated, not merely their centers. For a free scalar field, microcausality gives

$$[\hat{\Phi}_A, \hat{\Phi}_B] = i\Delta(F_A, F_B) = 0, \qquad W(F_A, F_B) = \langle 0|\hat{\Phi}_A\hat{\Phi}_B|0\rangle \text{ need not vanish}, \tag{S3}$$

where Δ is the Pauli–Jordan commutator distribution and $W(F_A, F_B)$ denotes the correspondingly smeared Wightman function. Thus the causal response between the regions vanishes while their vacuum correlations can remain nonzero. For initially ground-state two-level probes, the relevant block of the partial transpose, in the basis $\{|eg\rangle, |ge\rangle\}$, is

$$(\rho_{AB}^{T_B})_{\{|eg\rangle,|ge\rangle\}} = \begin{pmatrix} P_A & \mathcal{M} \\ \mathcal{M}^* & P_B \end{pmatrix} + O(\lambda^4). \tag{S4}$$

Its potentially negative eigenvalue at second order is

$$\lambda_-^{(2)} = \frac{P_A + P_B - \sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2}}{2}. \tag{S5}$$

The corresponding leading negativity is

$$\mathcal{N}^{(2)} = \max\left\{0, \frac{\sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2} - (P_A + P_B)}{2}\right\}, \tag{S6}$$

up to higher-order corrections. Its positivity condition is $|\mathcal{M}| > \sqrt{P_A P_B}$; for identical probes, $P_A = P_B = P$, it reduces to $\mathcal{N}^{(2)} = \max\{0, |\mathcal{M}| - P\}$. The exact expression must be re-derived if the probe dimension, initial state, coupling, or perturbative order is changed.

This two-region comparison has three distinct outputs. First, QTP supplies joint detection-time probabilities, whose leading terms involve four-point unequal-time field correlators; for the free Gaussian vacuum these can be reduced by Wick's theorem, but they are still event statistics rather than a detector density matrix. Second, an explicit two-probe detector instrument supplies the final probe state and hence the negativity in Eq. (S6). Third, the positive-formalism

and local-S-matrix approaches test whether the corresponding localized operations and their composition are probabilistically consistent and causally admissible. A QTP coincidence covariance,

$$C_{AB} = p_{AB} - p_A p_B, \tag{S7}$$

must therefore not be identified with $\mathcal{N}$: a nonzero covariance is a correlation witness, not by itself an entanglement measure.

A useful numerical and conceptual comparison should report the following quantities separately: (i) the integrated single-probe excitation probability and its time-resolved distribution; (ii) the joint event probability and coincidence covariance; (iii) the final two-probe density matrix and its resource witness, when an explicit probe instrument is specified; (iv) a no-signalling residual obtained by varying the remote setting while holding the local setting fixed; and (v) the detector's dark-count probability, efficiency, and readout resolution. The same field state, compact supports, smearing functions, probe preparation, and coupling must be used in all calculations. A timelike or causally connected configuration should be included as a control, not mixed with the strictly spacelike benchmark.

The comparison can then be summarized as follows:

Framework	Primary output in the benchmark	Question it can answer directly
UDW / explicit probe instrument	Probe transition probabilities and final probe state	How much resource is transferred to the specified probe?
QTP	Time-resolved and joint event probabilities	When do events occur, and what multi-time correlations are recorded?
Positive formalism	Probabilities for localized probes and their spacetime composition	Are localized measurements positive, local, and compositional?
Local S matrix	Local operations and induced Kraus updates	Are the operations causally admissible and non-signalling?

This benchmark suggests a disciplined order for future comparisons. First match the single-probe probability after integrating over the detection time. Next match joint event statistics for two strictly spacelike supports. Only after an explicit probe instrument has been fixed should one compare the final detector density matrix and its entanglement or other resource witnesses. Finally, test causal composition and finite-apparatus reliability. This order prevents a field correlator, an event-probability hierarchy, a POVM effect, a quantum instrument, and a resource measure from being treated as if they were the same physical object.

The measurement frameworks above define idealized operations and outcome probabilities. Practical resource certification also depends on finite detector sensitivity, noise, postselection, and the control of switching and spatial support.

S2 Independent reconstruction of the de Sitter interference calculation

To check the interference mechanism quantitatively, we independently reconstructed the de Sitter calculation of Zambianco, Teixidó-Bonfill, and Martín-Martínez for a massless, conformally coupled scalar field in four spacetime dimensions [6]. The spatial smearing can be integrated analytically, leaving a controlled numerical problem in detector time. Our aim is to test the reported parameter dependence and relative-phase effect, not to claim a new interference mechanism.

We use comoving detector centres, $t_{\mathrm{A}} = 0$ and $t_{\mathrm{B}} = \Delta t$, identical Gaussian switchings, and

$$a(t) = e^{Ht}, \qquad \eta(t) = \frac{1 - e^{-Ht}}{H}, \qquad \Delta\eta(t, t') = \frac{e^{-Ht'} - e^{-Ht}}{H}. \tag{S8}$$

The switching functions are $\chi_{\rm A}(t) = e^{-t^2/T^2}$ and $\chi_{\rm B}(t) = e^{-(t-\Delta t)^2/T^2}$. The two detector models are distinguished by their comoving widths:

$$\sigma_j(t) = \sigma \quad \text{or} \quad \sigma_j(t) = \frac{\sigma}{a(t)}. \tag{S9}$$

The first profile expands with the cosmological flow, whereas the second keeps its proper size fixed.

For the spatially smeared cross kernel, define

$$\Sigma_{AB}(t,t') = \sqrt{\sigma_{\rm A}(t)^2 + \sigma_{\rm B}(t')^2}. \tag{S10}$$

The radial momentum integral gives

$$K_{AB}(t,t') = \frac{2\pi}{\Sigma_{AB} d}\left[I\!\left(\frac{\Delta\eta + d}{\Sigma_{AB}}\right) - I\!\left(\frac{\Delta\eta - d}{\Sigma_{AB}}\right)\right], \tag{S11}$$

where

$$I(z) = i\sqrt{\frac{\pi}{2}}e^{-z^2/2} + \sqrt{2}\,D\!\left(\frac{z}{\sqrt{2}}\right), \qquad D(z) = e^{-z^2}\int_0^z e^{y^2}\,dy \tag{S12}$$

is the Dawson function. The communication kernel is

$$K^{-}_{AB}(t,t') = \frac{2\pi i}{\Sigma_{AB} d}\left[h\!\left(\frac{\Delta\eta + d}{\Sigma_{AB}}\right) - h\!\left(\frac{\Delta\eta - d}{\Sigma_{AB}}\right)\right], \qquad h(z) = \sqrt{\frac{\pi}{2}}e^{-z^2/2}. \tag{S13}$$

The local kernel follows from the $d \to 0$ limit of Eq. (S11). Substitution into the detector response gives $\mathcal{L}_{jj}$, $\mathcal{M}$, and $\mathcal{M}^-$; the field-correlation contribution is then $\mathcal{M}^+ = \mathcal{M} - \mathcal{M}^-$. The total negativity is computed from the full $\mathcal{M}$, not by adding the two source-specific negativities.

A useful change of variables for the time-ordered nonlocal integral is

$$s = \frac{t + t'}{2}, \qquad u = t - t' \geq 0. \tag{S14}$$

In these variables,

$$\Delta\eta(s,u) = \frac{2e^{-Hs}}{H}\sinh\!\left(\frac{Hu}{2}\right), \tag{S15}$$

and, in units $T = 1$,

$$\begin{aligned}&\chi_{\rm A}(s + u/2)\chi_{\rm B}(s - u/2) + \chi_{\rm A}(s - u/2)\chi_{\rm B}(s + u/2)\\ &\quad = e^{-2(s-\Delta t/2)^2}\left[e^{-(u+\Delta t)^2/2} + e^{-(u-\Delta t)^2/2}\right].\end{aligned} \tag{S16}$$

The light-cone structure is most efficiently resolved by using $z = \Delta\eta$ as the integration variable:

$$u(s,z) = \frac{2}{H}\,\mathrm{arsinh}\left(\frac{He^{Hs}z}{2}\right), \qquad \frac{\partial u}{\partial z} = \frac{e^{Hs}}{\sqrt{1 + (He^{Hs}z/2)^2}}. \tag{S17}$$

The Minkowski limit is regular: $u = z$ as $H \to 0$. In the numerical calculation, the Gaussian integral over s is handled by Gauss–Hermite quadrature, while the remaining light-cone integral is mapped through $z = d\tan\theta$. For the local response of expanding profiles, the corresponding variable is $\Delta\eta/\Sigma_{jj}$. For fixed-proper-size profiles, $\Sigma_{jj}(s,u) = \sigma e^{-Hs}\sqrt{2\cosh(Hu)}$, so both $\Delta\eta/\Sigma_{jj}$ and $e^{-2Hs}K_{jj}$ are independent of s. Consequently, in this conformal model the local response is independent of the detector centre for identical switchings translated in cosmic time; direct integration at two different centres confirms this identity numerically. This transformation is important: a uniform grid in u can under-resolve the spatially smeared light-cone peaks when the scale factor changes appreciably during the switching.

We set $\Omega T = 6$, $d/T = 2$, and use $\lambda = 1$ for the numerical values below; all leading-order response coefficients and the leading-order negativity scale as λ^2. The delay is scanned over $\Delta t/T \in [-6, 6]$. A separate direct two-dimensional trapezoidal integration with step $0.005T$ was used as a cross-check at $HT = 0.2$, $\sigma/T = 0.1$, and $\Delta t = 0$. The relative differences between the transformed and direct calculations are 3.7×10^{-5} for $\mathcal{L}_{\mathrm{AA}}$, 5.8×10^{-9} for $\mathcal{M}^-$, and about 1.0×10^{-3} for $\mathcal{M}^+$. The remaining difference in $\mathcal{M}^+$ is consistent with the slower convergence of the smooth, nonlocal part under the direct rectangular grid.

Table S1 shows the central-delay results for detectors whose comoving width is fixed. As HT increases, both $|\mathcal{M}^+|$ and $|\mathcal{M}^-|$ grow, but the relative phase approaches π. Consequently, their sum becomes much smaller than either contribution separately. At $HT = 0.4$, for example, $|\mathcal{M}^+|$ and $|\mathcal{M}^-|$ are each about an order of magnitude larger than the total $|\mathcal{M}|$. The ratio $|\mathcal{M}|/(|\mathcal{M}^+| + |\mathcal{M}^-|)$ falls from about 0.81 in the Minkowski limit to about 0.052 at $HT = 0.4$. This is a direct numerical manifestation of destructive interference, not a disappearance of the separate field-correlation and communication amplitudes.

Table S1: Independent reconstruction at $\Delta t = 0$ for expanding detector profiles, with $\Omega T = 6$, $d/T = 2$, $\sigma/T = 0.1$, and $\lambda = 1$. The amplitude and negativity columns are in units of 10^{-10}.

HT	$\lvert\mathcal{M}^+\rvert$	$\lvert\mathcal{M}^-\rvert$	$\lvert\mathcal{M}\rvert$	$\mathcal{N}$	φ (rad)
0	3.8730	1.0584	4.0150	3.7036	1.5708
0.2	6.0280	3.3284	3.0010	2.6463	2.8479
0.4	27.5935	25.7659	2.7482	2.2206	3.0646

The full delay scans reproduce the main qualitative features reported in the published figures. For $\sigma/T = 0.1$, increasing HT makes the separate amplitudes grow rapidly while the total negativity remains small on the same scale, as shown in Fig. S1. For $\sigma/T = 1$, the maximum negativity on the 49-point delay grid changes only modestly for profiles that expand with the Universe, whereas it increases markedly for detectors that retain their proper size. The sampled maxima are summarized in Table S2. The relative phase is the direct coefficient of the interference term; plotting $\cos\varphi$ avoids the branch discontinuities of φ and makes the contrast between the two detector models explicit in Fig. S3.

Table S2: Maximum leading-order negativity on the sampled 49-point grid $\Delta t/T \in [-6, 6]$, for $\Omega T = 6$, $d/T = 2$, and $\sigma/T = 1$. Values are in units of 10^{-9}; they are grid maxima, not continuous optimizations.

HT	Comoving width fixed	Proper detector size fixed
0.1	0.2408	0.2635
0.2	0.2271	0.3052
0.3	0.2215	0.4067
0.4	0.2223	0.6080
0.5	0.2271	1.1784

These calculations support the physical interpretation of the published study, while also sharpening its scope. The result concerns detectors in causal contact, with Gaussian temporal and spatial profiles of noncompact support. It is therefore not a calculation of strictly spacelike entanglement harvesting. The field is specifically massless and conformally coupled, and the detector dynamics are treated at leading perturbative order. The comparison between expanding and fixed-proper-size profiles is consequently a model comparison, not a universal theorem about all localized probes in an expanding universe. In particular, the large values of $|\mathcal{M}^{\pm}|$ at rapid expansion should not be interpreted as a corresponding increase in harvested entanglement: the relative phase can cancel most of the total nonlocal amplitude.

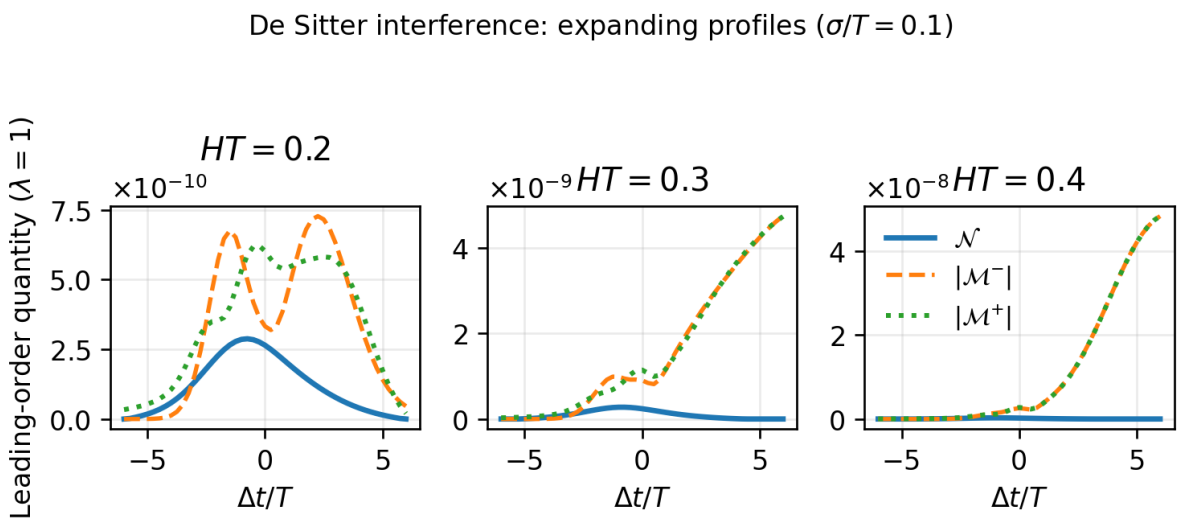


Figure S1: Independent reconstruction of the total negativity $\mathcal{N}$ and the source amplitudes $|\mathcal{M}^{\pm}|$ for expanding detector profiles, with $\Omega T = 6$, $d/T = 2$, and $\sigma/T = 0.1$. The three panels correspond to $HT = 0.2, 0.3, 0.4$. The different vertical scales are intentional: the separate amplitudes grow strongly with HT, while destructive interference keeps the total negativity comparatively small.

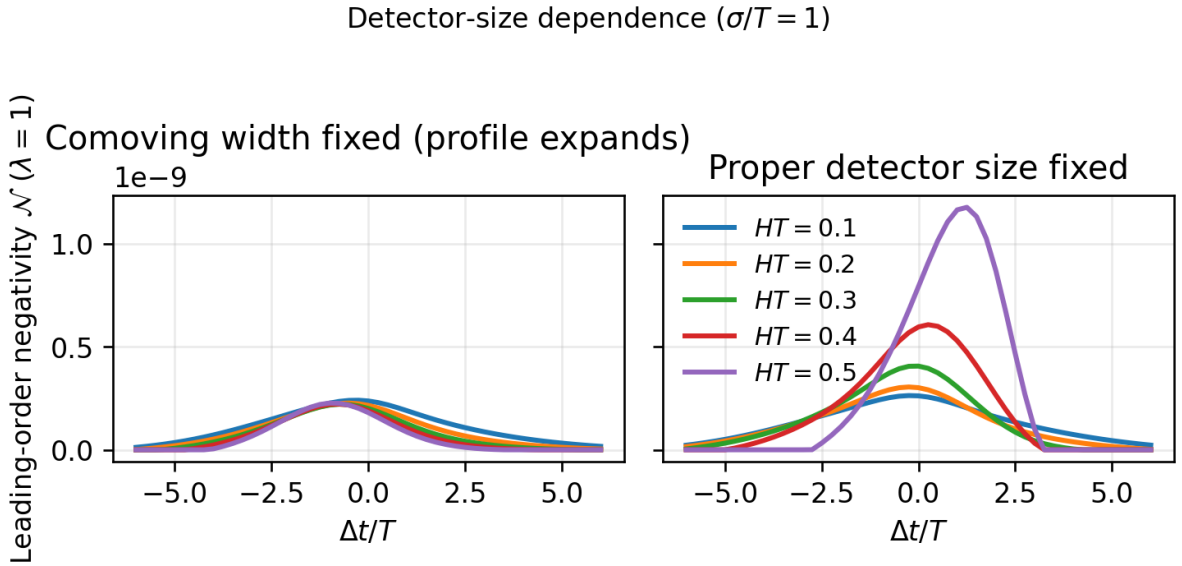


Figure S2: Negativity versus switching-time delay for $\sigma/T = 1$, comparing profiles with fixed comoving width and detectors with fixed proper size. Each panel contains $HT = 0.1, 0.2, 0.3, 0.4, 0.5$. The fixed-proper-size model becomes increasingly more effective at producing detector entanglement over the parameter range studied.

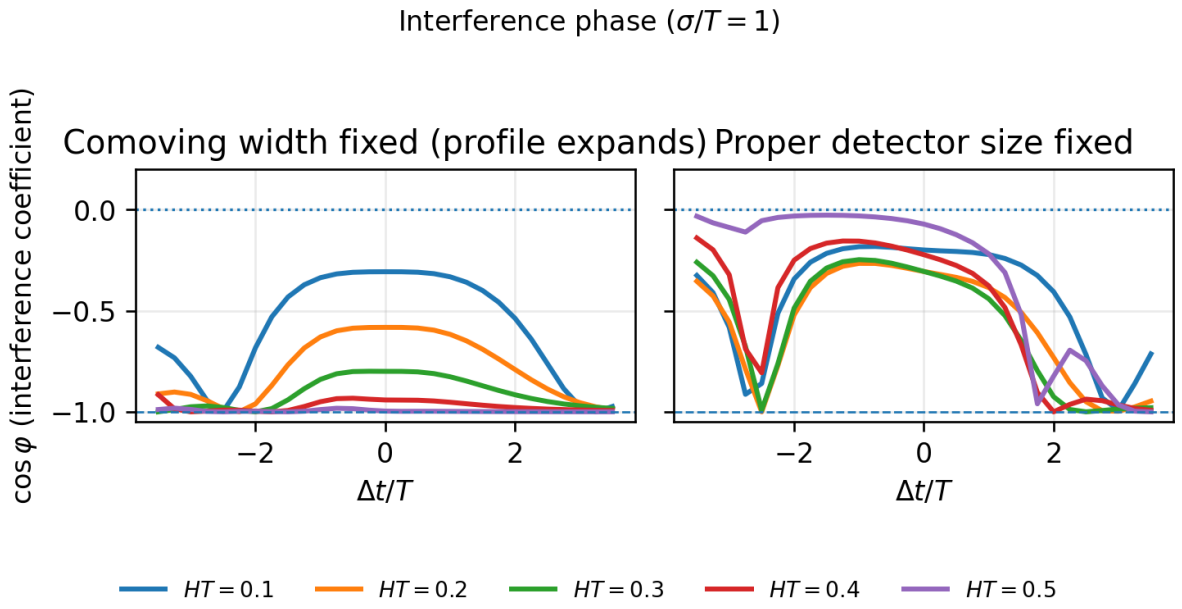


Figure S3: The interference coefficient $\cos\varphi$ for the two detector-size models at $\sigma/T = 1$. The phase is displayed only where both source amplitudes are numerically resolved. Negative values indicate destructive interference; values near zero correspond to approximate quadrature between the two contributions.

S3 Multipartite entanglement, topology, and perturbative control

Verification scope: the companion `verify_v7_core.py` independently constructs the biseparable counterexamples, checks the cluster-state channel argument, and reconstructs the first-order W and Dicke diagnostic model specified below. The source-specific Star crossover, second-order W matrix, and plotted reconstructions retained from the supplied material are not independently rederived from the primary articles in this revision. Their detailed source correspondence remains an author check; no conclusion here rules out revival in an exact finite-coupling model. Throughout this section $N = 2\mathcal{N}$, unlike the half-normalized negativity used in the main harvesting formulas.

For reproducibility, the first-order W/Dicke diagnostic uses the following explicit state model, with D the interacting qubit:

$$\widetilde{\rho}_q = \rho + \frac{\nu^2}{1-q}\left(\sigma_D^- \rho \sigma_D^+ + q\sigma_D^+ \rho \sigma_D^-\right), \qquad \rho_q^{\mathrm{test}} = \frac{\widetilde{\rho}_q}{\mathrm{Tr}\,\widetilde{\rho}_q}. \tag{S18}$$

The operators include the identity on the other qubits. The input is the equal superposition of all four-qubit computational states of Hamming weight one (W) or two (Dicke). This model reproduces the stated first-order normalization corrections and turning points. Its state-dependent normalization is not an exact CPTP evolution and cannot justify continuation to $q \to 1$. Source matching of this diagnostic to a particular published derivation is separate from the finite-matrix reproduction.

Recent work has examined whether the structure of an initial multipartite state can protect selected entanglement measures against acceleration. A useful example is the four-qubit cluster state

$$|\mathrm{CL}_4\rangle = \frac{1}{2}\big(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle\big), \tag{S19}$$

for which Li *et al.* report that the negativities across the $A|BCD$ and $B|ACD$ partitions remain maximal when detector D accelerates and interacts locally with a scalar field [7]. The result is partition specific: the partitions involving C or the accelerated detector D do not obey the same invariance.

For the $A|BCD$ and $B|ACD$ partitions, the freezing admits a stronger structural explanation than the first-order calculation alone suggests. Tracing detector D from Eq. (S19) already gives

$$N_{A|BC}(\rho_{ABC}) = N_{B|AC}(\rho_{ABC}) = 1, \qquad \rho_{ABC} = \mathrm{Tr}_D\, |\mathrm{CL}_4\rangle\langle \mathrm{CL}_4|, \tag{S20}$$

using the convention $N = \|\rho^{T_A}\|_1 - 1$. Suppose the field is initially uncorrelated with the detectors and only detector D couples to it. The exact reduced detector dynamics then defines a local completely positive trace-preserving map Λ_D on D. Trace preservation implies

$$\mathrm{Tr}_D[(I_{ABC} \otimes \Lambda_D)(\rho_{ABCD})] = \mathrm{Tr}_D\, \rho_{ABCD}. \tag{S21}$$

Partial trace is a local operation on the BCD side, so negativity monotonicity gives

$$N_{A|BCD}[(I_{ABC} \otimes \Lambda_D)(\rho)] \geq N_{A|BC}(\mathrm{Tr}_D\, \rho) = 1. \tag{S22}$$

Because the first subsystem is a qubit, the same negativity is bounded above by one. Hence

$$N_{A|BCD} = 1, \qquad N_{B|ACD} = 1, \tag{S23}$$

for *any* local CPTP channel on D, not only the perturbative Unruh–DeWitt channel. We verified this numerically for amplitude-damping and dephasing channels over their full parameter ranges. This observation strengthens the freezing statement but changes its interpretation: for these two partitions, the invariance is a state-and-channel structural property rather than evidence for a uniquely acceleration-induced protection mechanism.

The perturbative range remains relevant for other cluster-state quantities. In the published detector calculation, with $q = \exp(-2\pi\Omega/a)$ and effective coupling ν, the norm of the first-order state has the correction

$$\|\Psi\|^2 = 1 + \delta_{\mathrm{CL}}, \qquad \delta_{\mathrm{CL}} = \frac{\nu^2(1+q)}{2(1-q)}. \tag{S24}$$

At fixed nonzero ν^2, δ_{CL} is nonuniform as $q \to 1$. This does not undermine the exact argument in Eq. (S23), but it does limit conclusions about the detailed detector state, the $C|ABD$ and $D|ABC$ partitions, or task metrics derived from the truncated state in the infinite-acceleration limit.

A particularly useful test case for topology dependence is the asymmetric three-qubit Star state studied by Wu *et al.* [8]. The state has two peripheral nodes and one central node, so equal accelerations applied to structurally inequivalent subsystems need not give the same reduced dynamics. For acceleration of a peripheral node, the first-order normalization correction obtained from the published reduced state is

$$\delta_{\mathrm{per}} = \frac{\nu^2(1+3q)}{4(1-q)}, \tag{S25}$$

whereas for acceleration of the central node it is

$$\delta_{\mathrm{cen}} = \frac{\nu^2(1+q)}{2(1-q)}. \tag{S26}$$

Both corrections diverge as $q \to 1$ at fixed coupling, so high-acceleration behavior must be separated from parameter ranges in which the first-order expansion is quantitatively controlled.

We independently reconstructed the two published 8×8 reduced density matrices and recalculated their partial-transpose negativities. For $\nu^2 = 0.04$, the crossing of the two $N_{B(CA)}$ curves is reproduced at

$$q_\times = 0.238267\ldots, \tag{S27}$$

consistent with the reported $q \simeq 0.2383$. At the crossing, $\delta_{\mathrm{per}} \simeq 0.0225$ and $\delta_{\mathrm{cen}} \simeq 0.0325$, so this topology-dependent crossover occurs in a conservative perturbative regime. By contrast, the turning points preceding the reported high-q revival occur near $q \simeq 0.994$ and $q \simeq 0.991$, where the peripheral correction is approximately 6.4 and 4.6. For $\nu^2 = 0.04$, the condition $\delta < 0.1$ gives $q < 0.6923$ for peripheral acceleration and $q < 0.6667$ for central acceleration. Figure S4 shows the separation between the controlled crossover and the extreme-q turning point.

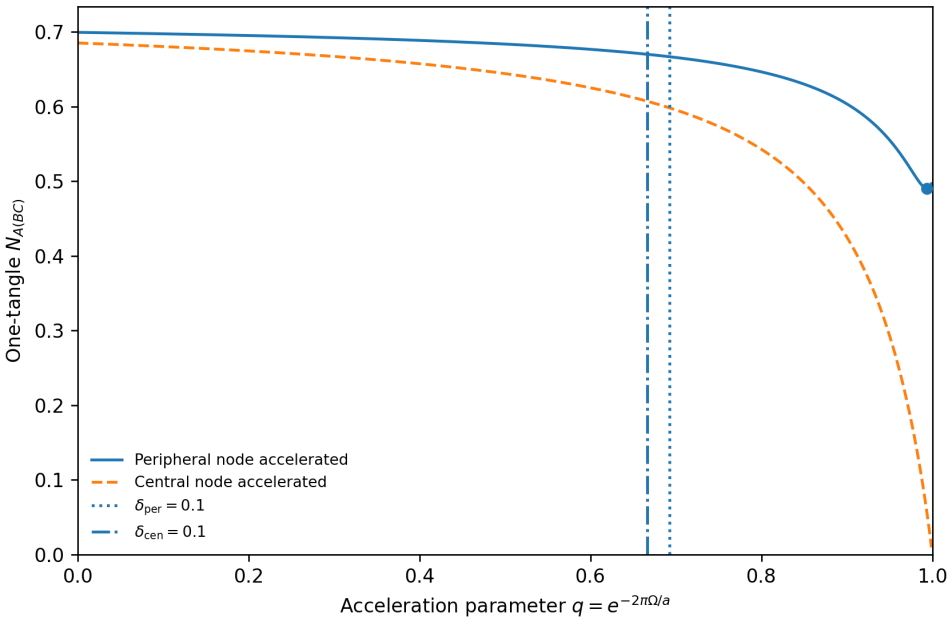


Figure S4: Independent reproduction of the Star-state one-tangle $N_{A(BC)}$ from the published reduced density matrices at $\nu^2 = 0.04$. The vertical markers indicate where the first-order normalization correction reaches $\delta = 0.1$. The high-q turning point lies far outside this conservative control region.

The same control test materially changes the interpretation of the reported W-state revival. For the four-qubit W state of Li *et al.* [9], the first-order correction is

$$\delta_W = \frac{\nu^2(1+3q)}{4(1-q)}. \tag{S28}$$

At $\nu^2 = 0.04$, our reconstruction gives the minimum of $N_{A(BCD)}$ at $q \simeq 0.982236$, where $\delta_W \simeq 2.22$. By comparison, $\delta_W < 0.1$ requires $q < 0.6923$, and even the loose condition $\delta_W < 1$ requires $q < 0.9612$. The paper also gives a second-order Dyson expansion. Reconstructing its Eqs. (39)–(41), we find the second-order minimum near $q \simeq 0.94427$, but there the natural ratio $\nu^2/(1-q) \simeq 0.718$ is still not small. Moreover, several displayed second-order coefficients change sign near $q \simeq 0.96$, and the truncated matrix develops a small negative eigenvalue of order 2.4×10^{-4} near the minimum. This is compatible with an omitted $O(\nu^4)$ correction, but it shows that the second-order truncation is not a uniformly positive density operator in the high-q region. Consequently, the non-monotonic trend is a valid feature of the published truncated

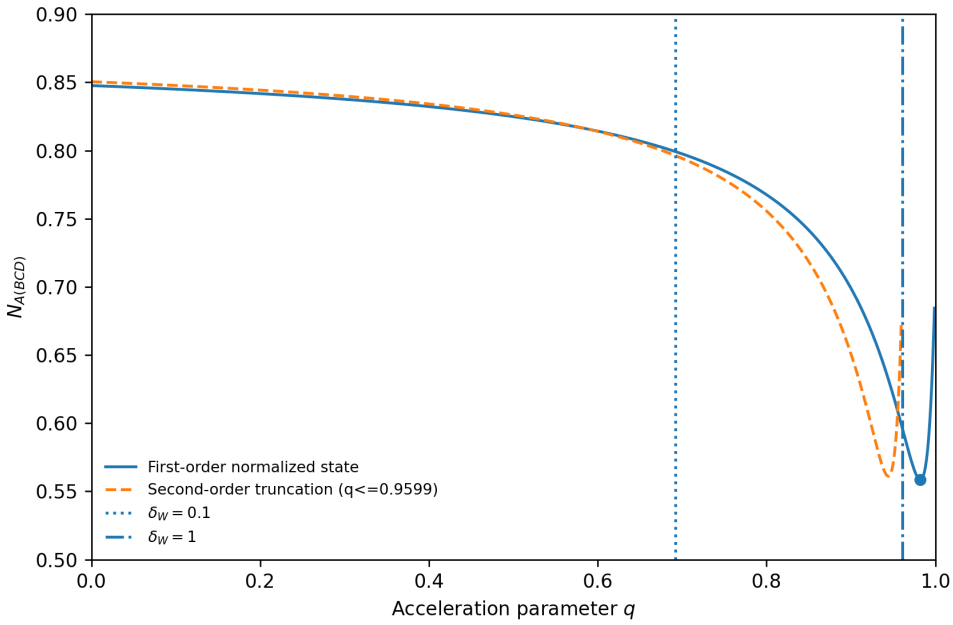


Figure S5: Independent W-state check at $\nu^2 = 0.04$. The first-order and second-order truncated predictions both display a minimum of $N_{A(BCD)}$, but the minima occur after the perturbative correction has ceased to be small. The second-order curve is shown only as an approximation diagnostic; its truncated matrix is not positive throughout the extreme-q region.

formulas, whereas a controlled physical revival toward $q \to 1$ remains unestablished by those truncations alone.

The Dicke-state revival shows the same issue even though the coupling used in that work is smaller. Wu *et al.* consider a four-qubit Dicke state with $\nu^2 = 0.01$ and report non-monotonic behavior of both $N_{A(BCD)}$ and the residual quantity π_4 [10]. Their first-order normalization correction is

$$\delta_D = \frac{\nu^2(1+q)}{2(1-q)}. \tag{S29}$$

Reconstructing the published reduced density matrix, we obtain the minimum of $N_{A(BCD)}$ at $q \simeq 0.992912$ and the minimum of the reported π_4 at essentially the same point. There $\delta_D \simeq 1.406$. The conservative requirement $\delta_D < 0.1$ gives $q < 0.9048$, while $\delta_D < 1$ gives $q < 0.99005$; at the value $q = 0.99$ used in the paper's coupling scan, $\delta_D \simeq 0.995$. Thus the upward branch begins only once the first-order correction is already order unity.

Mixed-state genuine multipartite entanglement requires an independent criterion

The approximation issue is logically separate from the definition of genuine multipartite entanglement (GME). A mixed state is biseparable if it can be written as a convex mixture of states that are separable across possibly different bipartitions; GME means that no such decomposition exists. This convex definition is why mixed-state certification cannot in general be inferred from a residual of pairwise negativities alone. General separability criteria and witnesses have been developed specifically for GME [11], and reviews of multipartite measures emphasize monotonicity, mixed-state extensions, and operational meaning as independent requirements [12].

For the three-qubit Star calculation, residual-negativity constructions are historically motivated by negativity monogamy [13], but the raw minimum residual is not a faithful GME

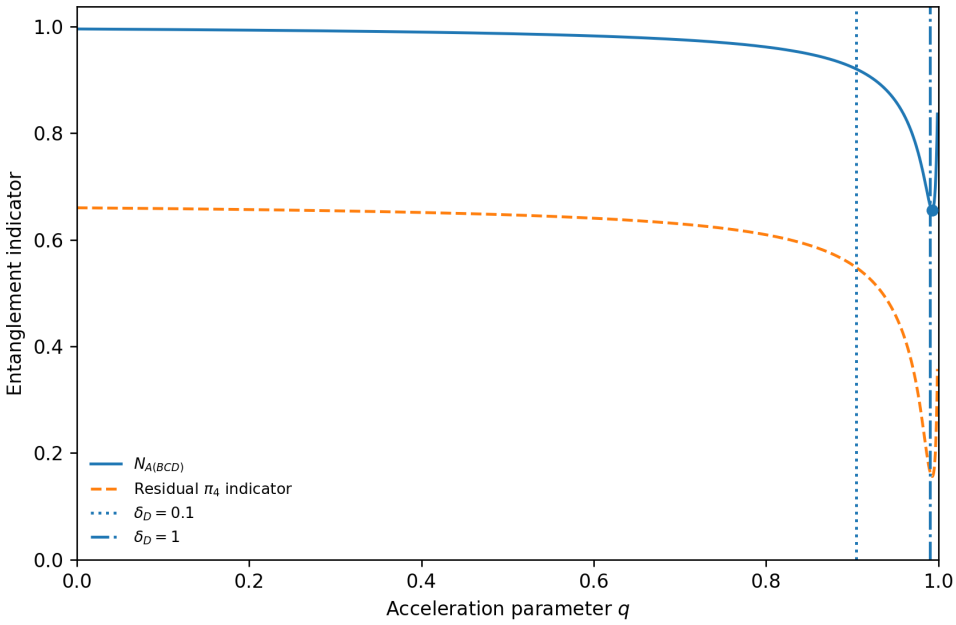


Figure S6: Independent Dicke-state reconstruction at $\nu^2 = 0.01$. Both $N_{A(BCD)}$ and the residual π_4 indicator turn upward only close to the region where the first-order normalization correction reaches order unity.

certificate on arbitrary mixed states. The explicitly biseparable state

$$\rho_{\text{bisep}} = \frac{1}{2}|\Phi^+\rangle_{AB}\langle\Phi^+| \otimes |0\rangle_C\langle 0| + \frac{1}{2}|\Phi^+\rangle_{AC}\langle\Phi^+| \otimes |0\rangle_B\langle 0|, \tag{S30}$$

has residual squared negativities approximately 0.3090, 0.1545, and 0.1545, despite being biseparable by construction. We therefore call this quantity a *minimum residual-negativity indicator* unless an independent mixed-state GME criterion is supplied.

The four-qubit Dicke residual π_4 has an analogous and even sharper limitation. Consider

$$\begin{aligned} \rho^{(4)}_{\text{bisep}} &= \frac{1}{3}\left(\rho_{AB|CD} + \rho_{AC|BD} + \rho_{AD|BC}\right), \\ \rho_{XY|ZW} &= \left(|\Phi^+\rangle\langle\Phi^+|\right)_{XY} \otimes \left(|\Phi^+\rangle\langle\Phi^+|\right)_{ZW}. \end{aligned} \tag{S31}$$

Each component is separable across a $2|2$ bipartition, so the mixture is biseparable. Nevertheless all four one-versus-three negativities equal one, all six two-qubit negativities vanish, and the residual definition used in the Dicke analysis gives

$$\pi_A = \pi_B = \pi_C = \pi_D = \pi_4 = 1. \tag{S32}$$

Thus a maximal raw π_4 does not, by itself, certify four-partite GME for an arbitrary mixed state. In this review we therefore use the term *residual* π_4 *indicator* when discussing the field-traced detector state, unless a separate GME witness, convex-roof construction, or proven mixed-state monotone is provided. Recent constructions such as the minimum entanglement drop illustrate the continuing effort to define multipartite monotones with explicit structural properties [14].

These checks make the comparison across initial states more precise. The W and Dicke calculations both contain an upward branch in the published truncated formulas, but in the parameter choices used for the plots the minima occur outside a small-correction regime. The Star-state topology crossover, in contrast, occurs at moderate q where the first-order correction is small. The cluster-state $A|BCD$ and $B|ACD$ freezing is stronger still: for local noise acting

only on D, it follows exactly from the initial state's reduced entanglement and local-channel structure, independently of the Unruh perturbation. None of these statements is directly comparable to the asymptotic open-system study of GHZ-type detectors by Berrada and Bougouffa, which uses genuine multipartite concurrence, tripartite mutual information, and purity after equilibration [15].

Table S3: Representative multipartite relativistic-resource studies and the distinctions needed for comparison.

Initial state	Regime	Main quantity	Reported behavior	Audit conclusion
Four-qubit W [9]	Finite-time UDW dynamics; first- and second-order formulas	$N_{A(BCD)}$	Decrease followed by recovery	Reproduced from published matrices; first-order minimum at $q \simeq 0.9822$ has $\delta_W \simeq 2.22$; second-order minimum also lacks a small expansion ratio.
Four-qubit cluster [7]	Local channel on accelerated detector D	Selected $1\|3$ negativities	$A\|BCD$ and $B\|ACD$ remain maximal	Exact for arbitrary local CPTP noise on D because $\mathrm{Tr}_D\,\rho$ already has maximal $A\|BC$ and $B\|AC$ negativity; not a uniquely Unruh-specific mechanism.
Three-qubit Star [8]	Finite-time first-order UDW dynamics	One-tangles; residual-negativity indicator	Topology-dependent crossover; high-q revival	Moderate-q crossover reproduced within control; high-q revival outside small-correction regime; residual is not a general mixed-state GME certificate.
Four-qubit Dicke [10]	Finite-time first-order UDW dynamics	$N_{A(BCD)}$; residual π_4 indicator	Non-monotonic decrease and increase	Minima near $q \simeq 0.9929$ have $\delta_D \simeq 1.41$; raw π_4 can equal one on an explicit biseparable mixed state.
GHZ-type [15]	Asymptotic detector state after equilibration	Genuine multipartite concurrence; mutual information; purity	Strong suppression of steady-state GME	Different dynamical regime and different resource definition; not a direct test of the finite-time revival claims.

The resulting synthesis is more specific than the statement that "acceleration degrades entanglement." Relativistic dynamics can produce qualitatively different curves for different partitions and measures, but a claimed "freezing," "revival," or "sudden death" must be classified according to its mechanism and evidence level. A freezing can be an exact local-channel invariant, a revival can occur only beyond the controlled range of a truncation, and a positive

residual can fail to certify GME on mixed states. Task-level claims require an additional step: channel capacity, teleportation fidelity, or computational utility must be demonstrated using the accessible subsystems and allowed operations of the protocol.

A recurring methodological error is to infer a hierarchy between resources from plots that use different normalizations or different parameter regimes. A larger numerical value of one measure than another has no general meaning unless the measures are defined on a common operational scale. The more robust comparison asks whether each resource is present, whether it can be certified with the available measurements, whether the approximation is controlled, and whether the resource enables a specified task under the same constraints.

S4 Transverse polarization kernel and orientation dependence

For the free electromagnetic vacuum, the electric-field two-point function contains the transverse projector

$$P_{ij}(\hat{\mathbf{k}}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \tag{S33}$$

which is the polarization sum entering the dipole-coupling calculation of Pozas-Kerstjens and Martín-Martínez [16]. Before detector-specific form factors are inserted, its angular integral already has the universal decomposition

$$I_{ij}(z) = A_\perp(z)(\delta_{ij} - \hat{R}_i \hat{R}_j) + A_\parallel(z)\hat{R}_i \hat{R}_j, \qquad z = kR, \tag{S34}$$

with

$$A_\perp(z) = \frac{4\pi}{3}[2j_0(z) - j_2(z)], \qquad A_\parallel(z) = \frac{8\pi}{3}[j_0(z) + j_2(z)]. \tag{S35}$$

At $z = 0$ both equal $8\pi/3$, so the local angular average is isotropic. At finite separation they differ and change sign at different kR. Direct numerical angular integration reproduces these expressions to machine precision, as shown in Fig. S7.

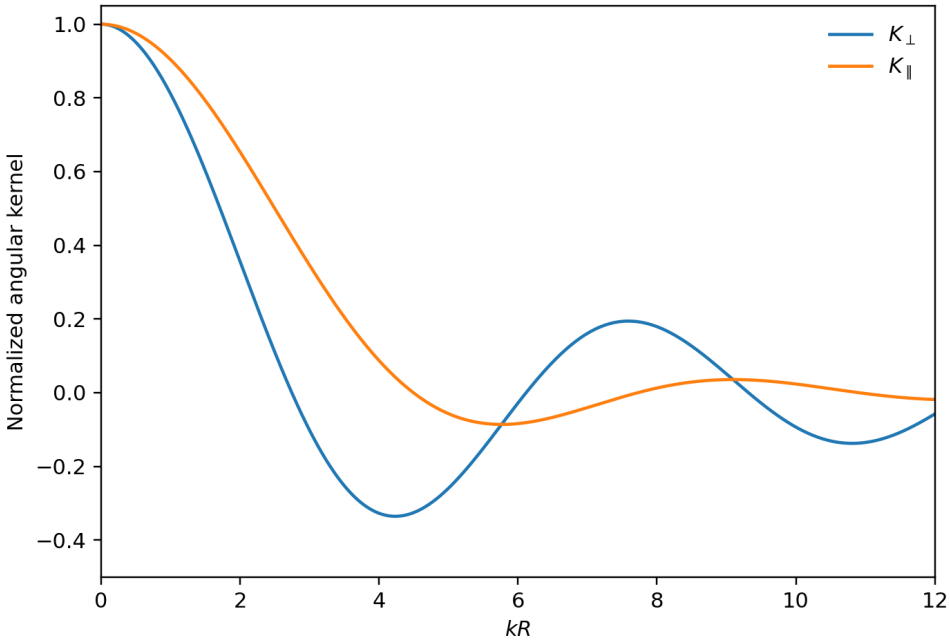


Figure S7: Universal transverse-polarization angular kernel before detector-specific radial or atomic form factors are included. The two contractions agree at zero separation but differ for $kR > 0$, showing how nonlocal electromagnetic correlations acquire orientation dependence.

Table S4: Finite-mode Jaynes–Cummings resource-transfer benchmark at $\theta = \pi/2$. Mutual informations are in nats.

r	I_{F}	$I(D_A : D_B)$	$\mathcal{N}$	Upper bound
0.05	0.03499	0.03480	0.04983	0.26453
0.10	0.11251	0.11009	0.09866	0.47436
0.20	0.34258	0.31445	0.18936	0.50000
0.40	0.96490	0.69518	0.32103	0.50000
0.60	1.68845	0.84182	0.36842	0.50000
0.80	2.45470	0.78027	0.34684	0.50000

This also clarifies a detector-specific caveat. In the hydrogenic calculation, vanishing harvesting for perpendicular $2p_z$ transition axes is a statement about that restricted transition channel; when the full degenerate $2p$ manifold is included, other angular-momentum channels can contribute [16]. Local isotropy therefore does not imply isotropic nonlocal harvesting, and field-vector structure should be separated from atomic selection rules.

S5 Finite-mode Jaynes–Cummings resource-transfer benchmark

A finite-mode model provides a transparent test of the field-to-detector bounds before one attempts a continuum-field calculation. Consider two bosonic modes in a two-mode squeezed vacuum,

$$|\Psi_r\rangle_{\mathrm{F}} = \sqrt{1-\lambda^2}\sum_{n=0}^{\infty}\lambda^n|n\rangle_A|n\rangle_B, \qquad \lambda = \tanh r, \tag{S36}$$

whose field mutual information is $I_{\mathrm{F}} = 2g(\sinh^2 r)$, with $g(n) = (n+1)\ln(n+1) - n\ln n$. Correlations between atoms and two-mode squeezed radiation have earlier been studied in generalized two-mode Jaynes–Cummings models; the present calculation specializes to two independent local resonant channels and evaluates the resulting detector state explicitly [17]. Each mode interacts locally with a two-level detector initially in its ground state. In the resonant Jaynes–Cummings model,

$$U_\nu(\theta) = \exp[-i\theta(\sigma_\nu^+ a_\nu + \sigma_\nu^- a_\nu^\dagger)], \qquad |g,n\rangle \mapsto \cos(\theta\sqrt{n})|g,n\rangle - i\sin(\theta\sqrt{n})|e,n-1\rangle, \tag{S37}$$

where the second term is absent for $n = 0$. Tracing out the field modes gives a two-qubit detector state. The Fock expansion is truncated at $N_{\max} = 30$; increasing the cutoff to 40 changes the reported quantities by less than the displayed precision over $0.05 \le r \le 0.8$. No postselection is used.

Figure S8 shows that the detector negativity depends nontrivially on the interaction angle. At $\theta = \pi/2$, representative values of the field and detector mutual informations and the detector negativity are listed in Table S4. They satisfy $I(D_A : D_B) \le I_{\mathrm{F}}$ and the two-qubit negativity bound. The table and angle scan are complementary: the former gives a fixed-angle comparison, while the latter shows how the transfer changes during the local interaction.

A sharper state-dependent estimate is obtained by applying data processing after only the first local interaction. For the normalized truncated Fock coefficients c_n, the state of detector D_A and the untouched mode F_B is

$$\rho_{D_A F_B} = \sum_{m=0}^{N_{\max}} |v_m\rangle\langle v_m|, \quad |v_m\rangle = c_m\cos(\theta\sqrt{m})|g,m\rangle - ic_{m+1}\sin(\theta\sqrt{m+1})|e,m+1\rangle, \tag{S38}$$

with $c_{N_{\max}+1} = 0$. Its mutual information is finite because D_A is a qubit. The one-sided bound in Eq. (S98) reduces the field-derived negativity bound by about 29% at $r = 0.05$ and 0.10

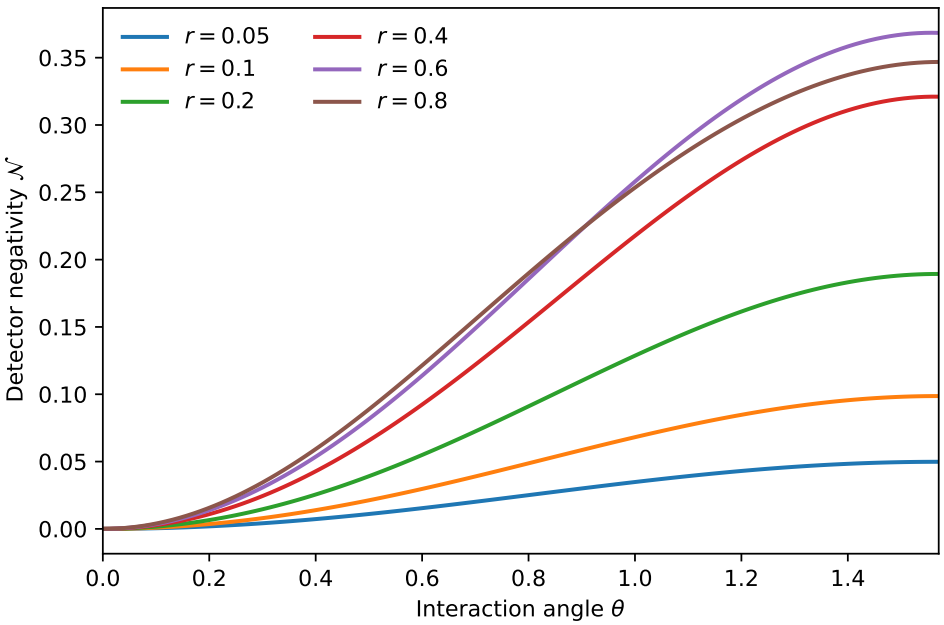


Figure S8: Detector negativity versus the local Jaynes–Cummings interaction angle for several initial squeezing parameters. Dashed lines show the two-qubit upper bound derived from the initial field mutual information. This finite-mode benchmark is not a spacelike-separated relativistic detector calculation.

for $\theta = \pi/4$; at larger squeezing the two-qubit cap $1/2$ becomes active. This is a state-specific estimate and does not contradict the unit worst-case contraction coefficient of the unbounded JC channel.

For the two-mode squeezed input, the detector state has an exact X form. Define

$$a_n = \cos^2(\theta\sqrt{n}), \qquad b_n = \sin^2(\theta\sqrt{n}). \tag{S39}$$

In the basis $(|gg\rangle, |ge\rangle, |eg\rangle, |ee\rangle)$,

$$\rho_{D_A D_B} = \begin{pmatrix} P_{gg} & 0 & 0 & Z \\ 0 & P_{ge} & 0 & 0 \\ 0 & 0 & P_{ge} & 0 \\ Z^* & 0 & 0 & P_{ee} \end{pmatrix}, \tag{S40}$$

where

$$\begin{aligned} P_{gg} &= (1-\lambda^2)\sum_{n=0}^{\infty} \lambda^{2n} a_n^2, \\ P_{ge} &= (1-\lambda^2)\sum_{n=0}^{\infty} \lambda^{2n} a_n b_n, \\ P_{ee} &= (1-\lambda^2)\sum_{n=0}^{\infty} \lambda^{2n} b_n^2, \\ Z &= -(1-\lambda^2)\lambda \sum_{n=0}^{\infty} \lambda^{2n} a_n b_{n+1}. \end{aligned} \tag{S41}$$

The partial transpose gives the exact negativity

$$\mathcal{N} = \max\left\{0,\, (1-\lambda^2)\left[\lambda \sum_{n=0}^{\infty} \lambda^{2n} a_n b_{n+1} - \sum_{n=0}^{\infty} \lambda^{2n} a_n b_n\right]\right\}. \tag{S42}$$

For weak squeezing, $\lambda = \tanh r \ll 1$,

$$\mathcal{N} = r\sin^2\theta + O(r^2), \qquad I_{\mathrm{F}} = 2r^2[1 - \ln(r^2)] + O(r^4|\ln r|). \tag{S43}$$

The detector negativity is therefore linear in r at leading order, whereas the field mutual information is of order $r^2|\ln r|$. This explains why the general mutual-information bound is loose for weakly squeezed inputs.

There is also a bound based only on local detector populations. For any two-qubit state,

$$\mathcal{N}(\rho_{D_AD_B}) \le \sqrt{\det \rho_{D_A}}, \qquad \rho_{D_A} = \begin{pmatrix} 1-p_A & c_A \\ c_A^* & p_A \end{pmatrix}. \tag{S44}$$

The JC interaction conserves $\hat{n}_A + |e_A\rangle\langle e_A|$, so $p_A \le \bar{n}_A$ and similarly for B. Hence

$$\mathcal{N} \le \min\{\nu(\bar{n}_A), \nu(\bar{n}_B)\}, \qquad \nu(x) = \sqrt{x_*(1-x_*)}, \quad x_* = \min\{x, \tfrac{1}{2}\}. \tag{S45}$$

This population bound is accessible from local measurements, but its excitation-conservation premise does not apply unchanged to the full Unruh–DeWitt interaction with counter-rotating terms.

The finite-mode benchmark confirms the internal consistency of the data-processing bounds and illustrates the dependence of resource transfer on the detector coupling. It is not a spacelike-separated relativistic harvesting calculation. A continuum-field extension must specify compact spacetime supports for the local modes and control the corresponding local-algebra mutual information.

S6 Strictly spacelike pointlike detector limit

We now examine whether the entanglement extracted by two detectors survives when their spatial size tends to zero while their separation and switching duration remain fixed. Consider a massless real scalar field in four-dimensional Minkowski spacetime. The two identical UDW detectors are at rest, their centres are separated by L, and their spatial profiles are normalized, spherically symmetric functions $F_\sigma(\mathbf{x})$ supported in balls of radius σ. The switching is

$$\chi(t) = \begin{cases} (1 - t^2/T^2)^6, & |t| \le T, \\ 0, & |t| > T, \end{cases} \qquad \widetilde{\chi}(\omega) = \int_{-\infty}^{\infty} dt\, \chi(t) e^{i\omega t}. \tag{S46}$$

This switching is compactly supported and has five continuous derivatives at the endpoints. We use the smooth compact spatial profile

$$F_\sigma(\mathbf{x}) = C_\sigma \exp\left[-\frac{1}{1 - |\mathbf{x}|^2/\sigma^2}\right] \quad (|\mathbf{x}| < \sigma), \tag{S47}$$

and set it to zero outside the ball; C_σ is fixed by $\int d^3x\, F_\sigma(\mathbf{x}) = 1$. The full interaction regions are strictly spacelike separated provided

$$L - 2\sigma > 2T. \tag{S48}$$

At leading order in the detector–field coupling λ, the local excitation probability and the nonlocal coherence are

$$P_\sigma = \frac{\lambda^2}{4\pi^2} \int_0^\infty dk\, k\, |\widetilde{F}_\sigma(k)|^2 |\widetilde{\chi}(\Omega + k)|^2, \tag{S49}$$

$$\mathcal{M}_\sigma = -\frac{\lambda^2}{4\pi^2 L} \int_0^\infty dk\, \sin(kL) |\widetilde{F}_\sigma(k)|^2 \widetilde{\chi}(\Omega - k) \widetilde{\chi}(\Omega + k), \tag{S50}$$

where $\widetilde{F}_\sigma(k)$ is the radial Fourier transform of the spatial profile. Because the two supports are strictly spacelike separated, the Wightman function is smooth on the product of the two interaction regions, and the nonlocal integral has no light-cone singularity. The local response is also finite in the pointlike limit because the compact switching has sufficient differentiability for $\widetilde{\chi}(\omega)$ to decay rapidly. The leading-order negativity for identical detectors is

$$\mathcal{N}_\sigma = \max\{0, |\mathcal{M}_\sigma| - P_\sigma\} + O(\lambda^4). \tag{S51}$$

The small-σ expansion follows from the second spatial moment. For a normalized isotropic profile,

$$\widetilde{F}_\sigma(k) = 1 - \frac{k^2\langle r^2\rangle_\sigma}{6} + O(k^4\sigma^4), \qquad \langle r^2\rangle_\sigma = \mu_2\sigma^2. \tag{S52}$$

Consequently, provided the switching makes the corresponding spectral moments convergent,

$$\frac{P_\sigma}{\lambda^2} = p_0 - \frac{\mu_2\sigma^2}{12\pi^2}\int_0^\infty dk\, k^3|\widetilde{\chi}(\Omega+k)|^2 + O(\sigma^4), \tag{S53}$$

$$\frac{\mathcal{M}_\sigma}{\lambda^2} = m_0 + \frac{\mu_2\sigma^2}{12\pi^2 L}\int_0^\infty dk\, k^2 \sin(kL)\widetilde{\chi}(\Omega-k)\widetilde{\chi}(\Omega+k) + O(\sigma^4), \tag{S54}$$

where p_0 and m_0 denote the pointlike limits of P_σ/λ^2 and $\mathcal{M}_\sigma/\lambda^2$, respectively. Thus both detector response coefficients have finite pointlike limits for this switching class. If $|m_0| > p_0$, the leading-order negativity also has a strictly positive pointlike limit; if $|m_0| < p_0$, it vanishes in a neighbourhood of the pointlike limit. The equality case requires higher-order analysis.

For a concrete test, we set $T = 1$, $\Omega T = 9$, and $L/T = 2.033333\ldots$. The profile in Eq. (S47) has dimensionless second moment $\mu_2 = 0.33508696\ldots$. The spacelike margin is positive throughout the scan, including the largest radius $\sigma/T = 0.016$. The Fourier integrals were evaluated with a radial quadrature for the compact profile and a Gauss–Legendre quadrature for the switching transform. An independent reconstruction supplied in `reproduce_s6_fourier.py` instead uses the analytic transform $\widetilde{\chi}(\omega) = 92160\, j_6(|\omega|)/|\omega|^6$ for $T = 1$, with its regular limit at zero. Comparing spectral cutoffs $k_{\max} = 300,500$ and radial quadrature orders $96, 160$ changes the response coefficients by less than 2.5×10^{-17} and agrees with the archived scan to 1.6×10^{-17}. These differences are numerical convergence diagnostics, not certified error bounds. The pointlike values are

$$p_0 = 2.13325790 \times 10^{-6}, \qquad |m_0| = 3.57653182 \times 10^{-6}, \tag{S55}$$

so that

$$\lim_{\sigma\to 0}\frac{\mathcal{N}_\sigma}{\lambda^2} = 1.44327392 \times 10^{-6} + O(\lambda^2). \tag{S56}$$

The fitted small-size expansion is

$$\frac{|\mathcal{M}_\sigma| - P_\sigma}{\lambda^2} = 1.44327392 \times 10^{-6} + 1.96659 \times 10^{-6}\left(\frac{\sigma}{T}\right)^2 + O\big((\sigma/T)^4\big)\,. \tag{S57}$$

The positive limit is therefore not caused by a finite spatial radius: it remains after the compact spatial profile is taken to a point. The local response, nonlocal coherence, and leading negativity approach finite pointlike limits. The positive entangling gap is shown directly in Fig. S9; the separate response and negativity plots are omitted because their limiting values are already stated explicitly above.

This result answers only the detector-side pointlike-limit question for the specified massless scalar model and switching. It does not establish the behaviour of the mutual information between shrinking field regions. In particular, the detector interaction samples a spacetime world tube of fixed temporal duration, whereas the mutual information of two equal-time spatial balls refers to different local regions. These quantities must not be identified. A separate algebraic

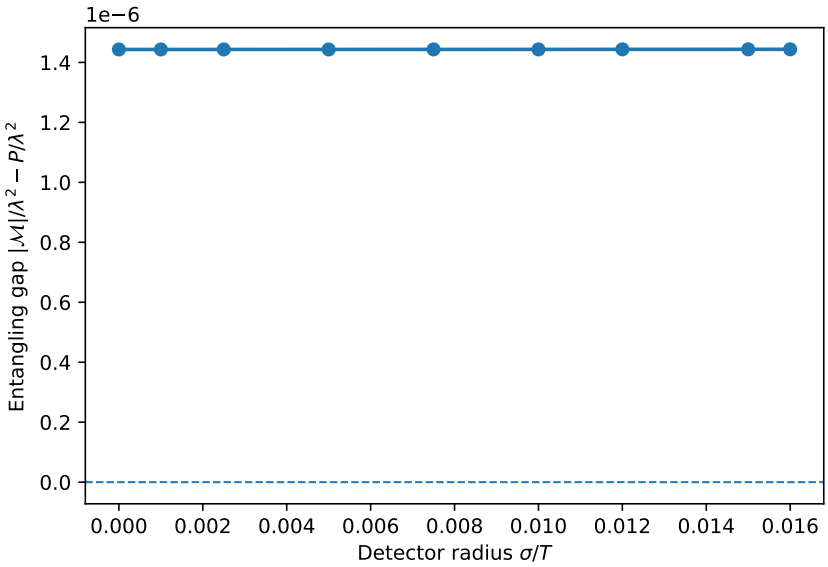


Figure S9: The entangling gap $|\mathcal{M}_\sigma|/\lambda^2 - P_\sigma/\lambda^2$. It remains positive throughout the strictly spacelike size scan.

or finite-resolution calculation is needed to determine the field-region mutual information and to compare its limit with the finite detector negativity. The present result also does not remove the need to keep the coupling perturbative: λ must be small enough that higher-order terms do not overturn the positive leading-order gap.

S7 Modular-flow estimate for the pointlike detector regions

This section collects the ball-region formulas used for the illustrative mutual-information estimate. They are not assumptions of the channel data-processing bound. For ball-shaped regions in a CFT vacuum, geometric modular flow can be written as follows; d denotes spacetime dimension, Δ a primary dimension, z the conformal cross ratio, $\dim \mathcal{R}$ the representation dimension, and k_d the strip coefficient. The two-copy expression and improved ansatz are taken within the scopes of Refs. [18, 19]. The ansatz has no certified four-dimensional error bound.

$$x_A^{\pm}(s) = R\,\frac{(R+x^{\pm}) - e^{\mp 2\pi s}(R-x^{\pm})}{(R+x^{\pm}) + e^{\mp 2\pi s}(R-x^{\pm})}, \tag{S58}$$

allowing replica correlators to be expressed through modular-flowed two-point functions. Agón, Casini, Gürsoy, and Planella Planas resum the two-copy sector into contributions labelled by CFT primaries; this improves the long-distance expansion but does not recover the short-distance area law by itself [18].

The construction has been extended to arbitrary-spin primaries and boosted balls [19]. For an unboosted ball and a scalar primary of dimension Δ, the two-copy term is

$$\mathcal{I}_\Delta^{(d)}(z) = \dim \mathcal{R} \int_0^1 dw\, \frac{(zw)^{2\Delta}\left(1+\sqrt{1-wz}\right)^{2(d-2\Delta)}}{4^{d+1}\sqrt{1-w}(1-wz)^{d/2}}, \tag{S59}$$

where d is the spacetime dimension, z the conformal cross ratio, and $\dim \mathcal{R}$ the Lorentz-representation dimension. The proposed interpolation

$$I_{\text{ansatz}}(z) = \mathcal{I}_\Delta^{(d-1)}(z) + \varsigma\, \mathcal{I}_{\Delta_*}^{(d-1)}(z), \tag{S60}$$

with ς fixed by the strip coefficient k_d,

$$\varsigma = k_d \frac{2^{d+3}\pi^{(d-2)/2}}{\dim \mathcal{R}\, \Gamma[(d-2)/2]} - 1, \qquad d > 2, \tag{S61}$$

For the pointlike detector benchmark, the enclosing double-cone radius is $R = T$ and the center separation is L_c. Let $D = L_c - 2R$ be the physical gap between the equal-time balls that are Cauchy surfaces of the double cones. The conformal cross ratio is

$$z = \frac{4R^2}{L_c^2} = \frac{4R^2}{(2R+D)^2}. \tag{S62}$$

Under the conformal map to concentric spheres of radii r_- and r_+, define $\bar{r} = (r_- + r_+)/2$ and $\delta_c = r_+ - r_-$. Their cross ratio obeys

$$1 - z = \left(\frac{\delta_c}{2\bar{r}}\right)^2. \tag{S63}$$

The short-distance expansion of the ordinary four-dimensional CFT mutual information is

$$I_{\mathrm{F}} \simeq 4\pi k_4 \left(\frac{\bar{r}}{\delta_c}\right)^2 = \frac{\pi k_4}{1-z}, \tag{S64}$$

where $k_4 = 0.0055351600$ is the strip coefficient for the four-dimensional free massless scalar [18, 19]. This is a leading asymptotic estimate, not an exact value at finite separation.

For $T = 1$, $R = 1$, and $L_c/T = 2.033333\ldots$, one has $D/T = 0.033333\ldots$ and $z = 0.96748186\ldots$. Equation (S64) gives

$$I_{\mathrm{F}}^{\mathrm{short}} \simeq 0.534754. \tag{S65}$$

We also evaluated the improved modular-flow ansatz in Eqs. (S60)–(S61) for the four-dimensional free scalar, taking $\Delta = 1$, $\dim \mathcal{R} = 1$, and k_4 as above. The coefficient is $\varsigma = 1.2258199\ldots$. Choosing the next scalar primary dimension $\Delta_* = 2$ gives

$$I_{\mathrm{F}}^{\mathrm{ansatz}} = 0.465718, \tag{S66}$$

while choosing $\Delta_* = 1.907890\ldots$ to remove the spurious subleading term gives $I_{\mathrm{F}}^{\mathrm{ansatz}} = 0.478713$. These two values are alternative model estimates, not rigorous error bars. The improved ansatz has been benchmarked against exact two-dimensional and numerical three-dimensional results, but an exact four-dimensional free-scalar curve at this cross ratio is not available for a direct error assessment.

For an exact field mutual information, the two-qubit bound is nontrivial only when $I_{\mathrm{F}} < 1/8$. The two near-contact ansatz estimates exceed this threshold and give the plug-in value $1/2$. That cap is independently valid by detector dimension; the estimates do not establish where the exact information-dependent bound becomes nontrivial.

A more separated scan suggests a window in which the ansatz plug-in value is below the dimensional cap while the leading-order UDW harvesting condition remains positive. Keeping the compact switching fixed and setting $\Omega T = 9$, the following table compares the modular-flow estimate with the pointlike detector response. The detector negativity column is the coefficient of λ^2 in $\mathcal{N} = \lambda^2 \max(0, |\mathcal{M}|/\lambda^2 - P/\lambda^2) + O(\lambda^4)$. Within this ansatz, $I_{\mathrm{F}} < 1/8$ for $L_c/T \gtrsim 2.10914$, while the leading-order detector gap remains positive up to $L_c/T \simeq 2.49030$. This is an overlap window of two model estimates, not a rigorous upper-bound interval: the ansatz is not known to upper-bound the exact field mutual information. Moreover, even at $L_c/T = 2.4$, the detector negativity coefficient is only 2.01×10^{-7}, whereas the ansatz plug-in

Table S5: Pointlike UDW detector scan at $\Omega T = 9$. The field mutual information is the improved modular-flow ansatz with $\Delta_* = 2$; it is an approximation, not an exact field-theory value.

L_c/T	$I_{\mathrm{F}}^{\mathrm{ansatz}}$	$\mathcal{N}_{\mathrm{LO}}/\lambda^2$	Ansatz plug-in
2.0333	0.46572	1.4433×10^{-6}	0.50000
2.2000	0.06050	7.7151×10^{-7}	0.34786
2.3000	0.03592	4.6121×10^{-7}	0.26803
2.4000	0.02421	2.0096×10^{-7}	0.22003
2.4500	0.02044	8.6134×10^{-8}	0.20221

value is about 0.22. The general Pinsker-based bound remains extremely loose and does not retain the weak-coupling scale.

The next theoretical improvement should therefore be a channel-sensitive bound. One possibility is to determine relative-entropy contraction coefficients for the actual local detector channels, rather than applying data processing only with coefficient one. Such a result could retain the weak-coupling and spectral-selectivity information that is lost when the complete field mutual information is used as the sole input.

S8 Derivative-coupled scalar detector response

$$\mathcal{F}_{\mathrm{der}}(\Omega) = \int d\tau\, d\tau'\, \chi(\tau)\chi(\tau') e^{-i\Omega(\tau-\tau')} \partial_\tau \partial_{\tau'} W_\phi\big(x(\tau), x(\tau')\big). \tag{S67}$$

If the switching and its derivatives vanish at the integration boundaries, integration by parts rewrites this as

$$\mathcal{F}_{\mathrm{der}}(\Omega) = \int d\tau\, d\tau'\, [\chi'(\tau) - i\Omega\chi(\tau)][\chi'(\tau') + i\Omega\chi(\tau')] e^{-i\Omega(\tau-\tau')} W_\phi\big(x(\tau), x(\tau')\big). \tag{S68}$$

The derivative model is therefore not the monopole model with the same switching function: it changes the frequency weighting and the effective temporal profile. In $(1+1)$ dimensions, differentiating a massless scalar two-point function also removes an additive infrared constant, which is one reason derivative coupling is useful in the reduced Schwarzschild models of Tjoa and Mann and of Gallock-Yoshimura *et al.* [20, 21]. This infrared advantage should not be mistaken for equivalence between the two interactions.

S9 Dirac-bilinear detector and charged-horizon qualifications

$$\mathcal{O}_\psi(x) =: \bar{\psi}(x)\psi(x) :, \qquad W_\psi(x,x') = \langle \mathcal{O}_\psi(x)\mathcal{O}_\psi(x')\rangle. \tag{S69}$$

The field itself obeys fermionic anticommutation relations at spacelike separation, while local even observables such as the scalar density obey the appropriate spacelike commutativity condition. Thus the detector can remain an ordinary two-level system with a bosonic monopole operator. The response is controlled by a composite correlator, not by replacing a scalar Bose factor with a Fermi factor by hand.

For the massless Dirac field in the conformally related $(1+1)$-dimensional setting used in that Schwarzschild study, Wick contraction gives the particularly useful identity

$$W_\psi^\alpha(x,x') = -N\, g^{\mu\rho}(x) \Lambda^\nu{}_\rho(x',x) \partial_\mu W_\phi^\alpha(x,x')\, \partial_{\nu'} W_\phi^\alpha(x,x'), \tag{S70}$$

where $N = 2$ is the spacetime dimension in that calculation and $\Lambda^\nu{}_\rho(x',x)$ is the vector parallel propagator. This relation is specific to the massless conformal model and its normal-ordering prescription; it is not a general identity between arbitrary Dirac and scalar detector responses.

It shows explicitly how spinor statistics and geometric parallel transport enter the correlator sampled by the detector, even when the final detector coupling is written using a scalar observable.

Dubey and Kolekar compare the Boulware, Hartle–Hawking–Israel, and Unruh states, as well as static and freely falling detector configurations [22]. In their covariant scalar-density coupling, the HHI static response has the expected Planckian behaviour, while the Unruh-state response is related to the Helmholtz free-energy density of a fermionic thermal bath. Their separate toy interaction that depends explicitly on a chosen 2-bein is not locally Lorentz invariant; its response can instead display bosonic or fermionic thermal expressions depending on the frame field used. These two interaction models must not be conflated. In the specified harvesting model, the anti-Hawking effect can enhance detector correlations, and nonzero mutual information and concurrence can persist near the horizon for static detectors, unlike the scalar-field comparison made in that work. The result is important evidence that changing the field and the detector operator can change the near-horizon conclusion, but it is not a universal theorem that fermionic fields always harvest more entanglement.

A complementary but operationally different line of work studies Dirac-mode entanglement across black-hole horizons. Chhieb *et al.* analyze fermionic mode splitting in Reissner–Nordström spacetime and report charge- and frequency-dependent redistribution of concurrence between exterior and interior modes [23]. This is a Bogoliubov-mode accessibility calculation, not a localized detector-harvesting protocol: its subsystem split is defined by the selected horizon regions and field modes, rather than by two finite spacetime-supported detector interactions. The distinction matters when comparing its "accessible entanglement" with the detector-state concurrence in the Schwarzschild harvesting calculation above. There is also a resource-definition caveat. The paper's reported Bures quantity is evaluated from the fidelity between the reduced state and the initial pure entangled state, rather than by minimizing Bures distance over the set of separable states. It therefore measures deviation from the initial state and should not automatically be treated as an entanglement monotone. This example reinforces the need to state the reference state and optimization set whenever a geometric distance is used as a resource indicator. A further charge-sector qualification is necessary. The paper writes a $U(1)$-gauge-coupled Dirac equation in the Reissner–Nordström potential, but its mode-mixing coefficients depend on ω/T_H alone and the field charge is not specified independently of the black-hole charge. For a genuinely charged field, the Hawking spectrum contains the horizon electrostatic chemical potential, so the thermal factor depends on $(\omega - q\Phi_H)/T_H$, where q is the field charge and Φ_H is the potential difference between the horizon and infinity [24]. Unless the model is explicitly neutral or its charge and gauge conventions remove this shift, the reported charge dependence should be read as a simplified mode-model result rather than a general prediction for a charged Dirac field.

S10 Massive scalar modes and a matched Schwarzschild detector benchmark

This section records the mode-level setup needed for a controlled massive-versus-massless comparison in four-dimensional Schwarzschild spacetime. It is a derivation of the benchmark and its spectral support; it does not claim that the two-detector numerical harvesting problem has already been solved.

S10.1 Radial equation and asymptotic channels

Using metric signature $(-,+,+,+)$, for a scalar field of mass m and curvature coupling ξ, the equation is $(\Box - m^2 - \xi R)\phi = 0$. In four-dimensional Schwarzschild spacetime $R = 0$, so ξ drops

out. With

$$\phi_{\omega\ell m_p}(t,r,\theta,\varphi) = e^{-i\omega t} Y_{\ell m_p}(\theta,\varphi)\frac{u_{\omega\ell}(r)}{r},$$

and tortoise coordinate $dr_*/dr = f^{-1}$, the radial equation is

$$\frac{d^2u_{\omega\ell}}{dr_*^2} + \left[\omega^2 - f(r)\left(m^2 + \frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3}\right)\right] u_{\omega\ell} = 0. \tag{S71}$$

The potential vanishes as $r_* \to -\infty$ at the horizon and tends to m^2 as $r_* \to +\infty$. This yields two independent continuum sectors. The 'in' sector is incident from spatial infinity and therefore has $\omega \geq m$, with asymptotic wave number $k_\infty = \sqrt{\omega^2 - m^2}$. The 'up' sector is incident from the past horizon and exists for every $\omega > 0$; for $0 < \omega < m$ its asymptotic tail is evanescent rather than propagating. These sub-threshold up modes remain part of the exterior field algebra and cannot be discarded from a local detector response.

The asymptotic problem is long ranged when $m > 0$. To make the boundary conditions explicit, set $r_h = 2M$, $x = r/r_h$, $y = r_*/r_h = x + \ln(x-1)$, $\mu = mr_h$, and $\epsilon = \omega r_h$. The dimensionless radial equation is

$$\frac{d^2u}{dy^2} + \left[\epsilon^2 - \left(1 - \frac{1}{x}\right)\left(\mu^2 + \frac{\ell(\ell+1)}{x^2} + \frac{1}{x^3}\right)\right] u = 0. \tag{S72}$$

Since $x = y - \ln y + O(\ln y/y)$, the potential has the more informative expansion

$$U_\ell(y) = \mu^2 - \frac{\mu^2}{y} + \frac{\ell(\ell+1) - \mu^2 \ln y}{y^2} + O\left(\frac{(\ln y)^2}{y^3}\right), \qquad y \to +\infty. \tag{S73}$$

The $1/y$ term is the long-range Coulomb tail. The angular term and the logarithmic correction generated by inverting the tortoise coordinate both enter at the next order; retaining one while silently dropping the other would give an inconsistent subleading approximation. For $\epsilon > \mu$, let $k = \sqrt{\epsilon^2 - \mu^2}$ and $\eta = \mu^2/(2k)$. The leading asymptotic phase is

$$\theta_\ell(y) = ky + \eta \ln(2ky) + \delta_\ell + o(1), \tag{S74}$$

where δ_ℓ contains the constant phase shift. Thus the incoming and outgoing waves are Coulomb-modified waves, not bare $e^{\mp iky}$ plane waves. For $0 < \epsilon < \mu$, retaining only the leading Coulomb tail gives the Whittaker solution

$$u^{(C)}_{\rm dec}(y) \propto W_{\lambda,1/2}(2\kappa y), \qquad \kappa = \sqrt{\mu^2 - \epsilon^2}, \qquad \lambda = \frac{\mu^2}{2\kappa}, \tag{S75}$$

whose large-y behaviour is $e^{-\kappa y}(2\kappa y)^\lambda$. This leading Coulomb solution does not include the $O(\ln y/y^2)$ and angular corrections in the full potential. For numerical matching we therefore use the full-potential leading WKB logarithmic derivative

$$\frac{u'_{\rm dec}(y)}{u_{\rm dec}(y)} = -q_\ell(y) - \frac{q'_\ell(y)}{2q_\ell(y)}, \qquad q_\ell(y) = \sqrt{U_\ell(y) - \epsilon^2}, \tag{S76}$$

at a large outer matching radius in the evanescent region. The threshold $\epsilon = \mu$ has a separate leading WKB behaviour, $u(y) \sim y^{1/4}\exp[\pm i2\mu\sqrt{y}]$, and is not obtained by substituting $k = 0$ into either asymptotic formula. These asymptotics are essential for stable frequency scans near $\omega = m$ and for avoiding a spurious phase or spectral-weight error from imposing short-range plane-wave boundary data [25, 26].

For clarity, the 'in' and 'up' modes below are normalized as scattering channels by the Klein–Gordon inner product. Let $\mathfrak{u}^{\rm in}_{\omega\ell}$ have unit incoming Coulomb-wave amplitude at spatial infinity,

and let $\mathfrak{u}^{\rm up}_{\omega\ell}$ have unit incoming amplitude at the horizon. In this convention the corresponding positive-frequency modes can be written as

$$v^{\rm in}_{\omega\ell m_p}(x) = \frac{e^{-i\omega t}Y_{\ell m_p}(\theta,\varphi)}{r\sqrt{4\pi\omega}}\sqrt{\frac{\omega}{k_\infty}}\,\mathfrak{u}^{\rm in}_{\omega\ell}(r), \tag{S77}$$

$$v^{\rm up}_{\omega\ell m_p}(x) = \frac{e^{-i\omega t}Y_{\ell m_p}(\theta,\varphi)}{r\sqrt{4\pi\omega}}\,\mathfrak{u}^{\rm up}_{\omega\ell}(r). \tag{S78}$$

The factor $\sqrt{\omega/k_\infty}$ in the 'in' mode is required by the Klein–Gordon normalization when the radial solution has unit incoming amplitude at infinity: $\delta(k_\infty - k'_\infty) = (k_\infty/\omega)\delta(\omega-\omega')$. It tends to one in the massless limit. The threshold limit must be taken after solving and normalizing the radial mode, rather than by substituting $k_\infty = 0$ into the prefactor. Consequently, the local spectral weights used below are, in this convention,

$$\rho^{\rm in}_{\omega\ell}(r_0) = \frac{1}{4\pi k_\infty r_0^2}|\mathfrak{u}^{\rm in}_{\omega\ell}(r_0)|^2, \qquad \rho^{\rm up}_{\omega\ell}(r_0) = \frac{1}{4\pi\omega r_0^2}|\mathfrak{u}^{\rm up}_{\omega\ell}(r_0)|^2. \tag{S79}$$

This makes the normalization convention behind $\rho^a_{\omega\ell}$ explicit. Other conventions are admissible, but the mode normalization, integration measure, and spectral density must be changed together.

For $\omega > m$, both scattering channels are open and span a two-dimensional degenerate mode space. They may be replaced by unitary linear combinations; a complementary canonical-quantization analysis constructs combinations with different far-field and near-horizon localization properties [26]. This basis freedom does not invalidate the 'in'–'up' basis. It does, however, require the occupation covariance matrix to be transformed with the modes. In the Unruh state, the occupation matrix in the 'in'–'up' basis is $N = \mathrm{diag}(0, n_H)$, where $n_H = (e^{\beta_H\omega}-1)^{-1}$. Under a balanced rotation $U = 2^{-1/2}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ it becomes

$$N' = UNU^\dagger = \frac{n_H}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}, \tag{S80}$$

so off-diagonal mode correlations appear. One must not rotate the modes and then assign independent diagonal thermal populations by hand. The 'in'–'up' basis is particularly convenient for the Unruh state because its incoming-vacuum and horizon-thermal sectors are diagonal in that basis.

S10.2 Stationary states and a static detector

Let $s \in \{\mathrm{in}, \mathrm{up}\}$ label the two sectors and let $\rho^s_{\omega\ell}(r_0)$ denote the positive local spectral density formed from the flux-normalized radial mode at the detector radius, including the factor r_0^{-2} and the chosen continuum normalization. For a static trajectory at r_0, proper time and Killing time are related by $\tau = \sqrt{f_0}t$, where $f_0 = f(r_0)$. After summing over the azimuthal index, the pulled-back Wightman function has the form

$$W_m(s) = \sum_{\ell=0}^{\infty}\sum_{a\in\{\mathrm{in},\mathrm{up}\}}\frac{2\ell+1}{4\pi}\int_{I_a} d\omega\,\rho^a_{\omega\ell}(r_0)\left[(1+n^a_\omega)e^{-i\omega s/\sqrt{f_0}} + n^a_\omega e^{+i\omega s/\sqrt{f_0}}\right], \tag{S81}$$

where $I_{\rm in} = [m,\infty)$ and $I_{\rm up} = (0,\infty)$. The occupation factors are state dependent:

$$\begin{aligned}
&\text{Boulware:} && n^{\rm in}_\omega = n^{\rm up}_\omega = 0,\\
&\text{Unruh:} && n^{\rm in}_\omega = 0, \quad n^{\rm up}_\omega = \frac{1}{e^{\beta_H\omega}-1},\\
&\text{Hartle–Hawking:} && n^{\rm in}_\omega = n^{\rm up}_\omega = \frac{1}{e^{\beta_H\omega}-1}, \qquad \beta_H = 8\pi M.
\end{aligned} \tag{S82}$$

For the stationary response convention $\dot{\mathcal{F}}_m(\Omega) = \int_{-\infty}^{\infty} ds\, e^{-i\Omega s} W_m(s)$, one obtains

$$\dot{\mathcal{F}}_m(\Omega) = 2\pi \sum_{\ell,a} \frac{2\ell+1}{4\pi} \int_{I_a} d\omega\, \rho^a_{\omega\ell}(r_0) \left[n^a_\omega \delta\!\left(\Omega - \frac{\omega}{\sqrt{f_0}}\right) + (1+n^a_\omega)\delta\!\left(\Omega + \frac{\omega}{\sqrt{f_0}}\right)\right]. \tag{S83}$$

For an excitation with $\Omega > 0$, only the n^a_ω term contributes. Hence the Unruh-state excitation samples thermally populated up modes at $\omega = \Omega\sqrt{f_0}$ even when this frequency is below m; the in sector is absent there. In the Hartle–Hawking state both sectors contribute where their spectral support permits, while the Boulware stationary excitation rate vanishes. Finite-time switching can add transient excitations and must be treated separately from this stationary rate. The Hartle–Hawking state in an asymptotically flat black-hole exterior is an equilibrium idealization with a thermal bath at infinity; for an evaporating black hole the Unruh state is the more relevant choice.

The Hartle–Hawking response also provides an exact thermal-consistency check. The detailed-balance ratio is insensitive to an overall spectral normalization, so it does not independently validate the radial amplitudes or their absolute normalization. Since both sectors have the same Bose occupation number, the response obeys

$$\dot{\mathcal{F}}_{\rm HH}(+\Omega) = e^{-\beta_H \Omega\sqrt{f_0}} \dot{\mathcal{F}}_{\rm HH}(-\Omega), \qquad \Omega > 0, \tag{S84}$$

which is detailed balance at the Tolman temperature $T_{\rm loc} = 1/(\beta_H\sqrt{f_0})$. The Unruh response does not obey this single-temperature relation in general because the 'in' and 'up' sectors have different occupations.

S10.3 Two-detector cross correlations

For two static detectors at radii r_A, r_B and angular separation γ, the azimuthally summed cross-correlator contains the angular factor $P_\ell(\cos\gamma)$ and the radial product of the corresponding in/up modes:

$$W_{AB}(t_A,t_B) = \sum_{\ell,a} \frac{2\ell+1}{4\pi} P_\ell(\cos\gamma) \int_{I_a} d\omega \Big[(1+n^a_\omega)\rho^a_{\omega\ell}(r_A,r_B) e^{-i\omega\Delta t} + n^a_\omega \rho^{a*}_{\omega\ell}(r_A,r_B) e^{+i\omega\Delta t}\Big], \qquad \Delta t = t_A - t_B. \tag{S85}$$

In the normalization of Eq. (S78), the cross spectral factors are

$$\rho^{\rm in}_{\omega\ell}(r_A,r_B) = \frac{1}{4\pi k_\infty r_A r_B} \mathfrak{u}^{\rm in}_{\omega\ell}(r_A)\mathfrak{u}^{\rm in*}_{\omega\ell}(r_B), \qquad \rho^{\rm up}_{\omega\ell}(r_A,r_B) = \frac{1}{4\pi\omega r_A r_B} \mathfrak{u}^{\rm up}_{\omega\ell}(r_A)\mathfrak{u}^{\rm up*}_{\omega\ell}(r_B). \tag{S86}$$

The complex conjugation in the second term of Eq. (S85) is required. Expanding a real field as $av + a^\dagger v^*$ gives $(1+n)v_A v_B^* + n v_A^* v_B$, not a common complex radial factor for both terms. For each mode the commutator contribution is consequently $C_{AB}e^{-i\omega\Delta t} - C^*_{AB}e^{+i\omega\Delta t}$ and is independent of n. Also $\rho^a_{\omega\ell}(r_B,r_A) = \rho^{a*}_{\omega\ell}(r_A,r_B)$, while $r_A = r_B$ gives the real local density used in S10.2. These are algebraic consistency checks; they do not establish convergence of the radial solver or a two-detector harvesting result. The accompanying `verify_v7_core.py` tests occupation-number cancellation using complex mode factors.

Unlike the local weights, these cross factors are generally complex. The local excitation probabilities and the time-ordered nonlocal matrix element must be computed from this same mode basis, with identical switching and spatial profiles in the massive and massless runs. A useful dimensionless parameter set is

$$(mM, \Omega_A M, \Omega_B M, r_A/M, r_B/M, \gamma, \Delta t/M, \sigma/M). \tag{S87}$$

Strict spacelike separation must be checked for the complete interaction world tubes, not only for the detector centres. The leading-order entanglement condition is then tested with the same partial-transpose criterion in both models.

S10.4 Radial-mode and local-response benchmark

We now give a numerical benchmark for the stationary response of one static detector. This calculation tests the radial scattering problem and the sub-threshold horizon-originating sector; it is not a two-detector harvesting calculation. We set $M = 1$, $r_0 = 4M$, so that $f_0 = 1/2$, and choose the proper detector gap $\Omega M = 0.2$. The sampled Killing frequency is therefore $\omega M = \Omega M\sqrt{f_0} = 0.1414213562$, with dimensionless frequency $\epsilon = 2M\omega = 0.2828427125$. The radial equation is solved independently in the in and up channels, and the angular sum is retained through $\ell = 10$.

Table S6 reports the dimensionless stationary rates $M\dot{\mathcal{F}}(+\Omega)$ and $M\dot{\mathcal{F}}(-\Omega)$. The Boulware excitation rate vanishes for this stationary, infinitely long interaction because the Boulware state is the ground state with respect to Schwarzschild time. In the Unruh state only the up channel is thermally occupied. For $mM = 0.15$, the sampled Killing frequency is below the mass threshold, so the in channel is closed while the up channel remains present. The nonzero excitation rate in this case is a direct numerical check that a mass threshold at spatial infinity is not a universal cutoff on the local Unruh response.

Table S6: Static-detector stationary response at $r_0 = 4M$ and $\Omega M = 0.2$. Rates are dimensionless, in units of M^{-1} before multiplication by M. The Hartle–Hawking columns give excitation and de-excitation rates.

mM	in channel	$M\dot{\mathcal{F}}_U(+\Omega)$	$M\dot{\mathcal{F}}_{HH}(+\Omega)$	$M\dot{\mathcal{F}}_{HH}(-\Omega)$
0	open	3.1348932×10^{-4}	1.0031788×10^{-3}	3.5074701×10^{-2}
0.05	open	3.1981180×10^{-4}	9.9265970×10^{-4}	3.4706916×10^{-2}
0.15	closed	5.1603148×10^{-4}	5.1603148×10^{-4}	1.8042297×10^{-2}

The Hartle–Hawking detailed-balance ratio agrees with $\exp[-\beta_H\Omega\sqrt{f_0}]$ to a relative numerical error below 3.4×10^{-16} for all three masses. Increasing the angular cutoff from $\ell = 6$ to $\ell = 10$ changes the summed excitation rates by less than 1.2×10^{-14} in the displayed parameter cases. For the sub-threshold up mode at $mM = 0.15$, changing the outer matching point from $y_{\max} = 250$ to 1500 changes the detector-radius mode amplitude by less than 2×10^{-12}; above threshold, the corresponding $\ell = 0$ amplitudes stabilize at the few-parts-in-10^7 level over $y_{\max} = 600$ to 1500. The flux-conservation residuals for the open channels are below 5×10^{-12}. At fixed Killing frequency, the $mM = 10^{-4}$ radial amplitudes approach the massless values with relative differences below 1.3×10^{-7} for the two $\ell = 0$ channels.

A further frequency scan at $mM = 0.1435$ reveals a real-frequency enhancement in the $\ell = 0$ Unruh response below the asymptotic mass threshold. The maximum occurs at $\Omega M = 0.20033355$ and gives $M\dot{\mathcal{F}}_U(+\Omega) = 4.4047823 \times 10^{-3}$, compared with 2.8756484×10^{-4} for the massless $\ell = 0$ response at the same proper gap. The ratio is approximately 15.32. The effective potential has three turning points at $x = r/(2M) = 1.10924$, 2.94418, and 38.88055; the detector at $x_0 = 2$ lies in the inner barrier, while the outer interval is a classically allowed well. The outer-well WKB action gives $S/\pi \simeq 0.569$ and is reported only as a diagnostic. The response peak is therefore described here as a *resonance-like local spectral enhancement*. We have not computed a complex-frequency pole or established a quasi-bound-state spectrum, so neither claim is made.

S10.5 Two-detector calculation still required

The local-response benchmark does not determine the two-detector resource. A complete harvesting calculation must use the complex cross-spectral factors in Eq. (S86), retain their radial phases, and evaluate the time-ordered nonlocal matrix element with the same mode normalization as the local terms. It must then impose identical compact switching and spatial profiles in the massive and massless cases, verify strict spacelike separation of the complete interaction

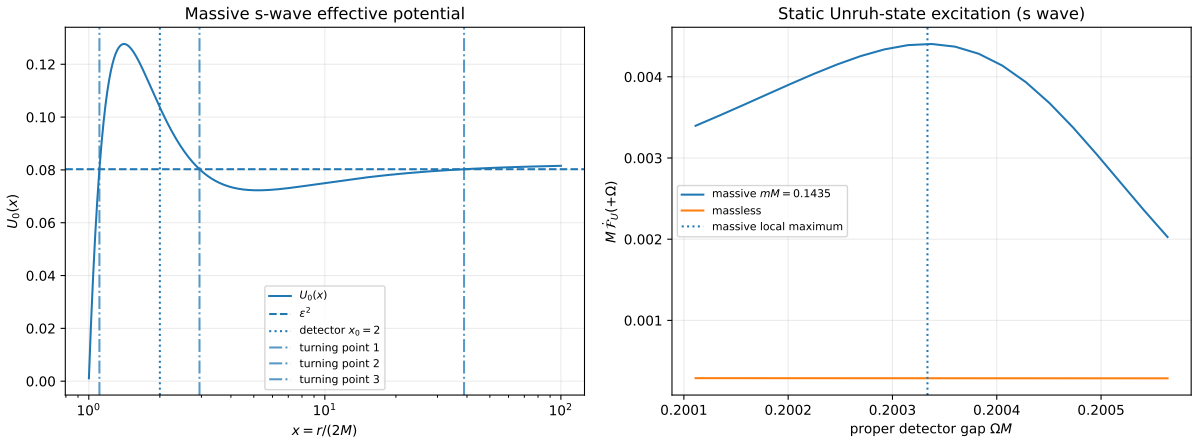


Figure S10: Massive Schwarzschild s-wave benchmark at $mM = 0.1435$. Left: effective potential, sampled energy, detector radius, and three turning points. Right: real-frequency Unruh excitation rate for a static detector, compared with the massless s-wave at matched proper gap. The vertical dotted line marks the numerically optimized local maximum. This is a single-detector response, not a harvesting negativity or a computed complex pole.

world tubes, converge the frequency and angular-momentum integrations, and compare the leading-order negativity and $|\mathcal{M}|/\sqrt{P_A P_B}$ at matched dimensionless parameters. Until that calculation is completed, the local spectral enhancement found above cannot be interpreted as enhanced harvested entanglement, and no monotonicity claim about the effect of m on harvesting is justified.

S11 Channel contraction: assumptions and counterexamples

A strict data-processing contraction can sharpen the field-to-detector bound, but it must be established for the actual detector channel and input class. For a regulated channel $\mathcal{E} : A \to D$, define

$$\eta_{\mathrm{MI}}(\mathcal{E}) = \sup_{\rho_{RA}: I(R:A)>0} \frac{I(R:D)_{(\mathrm{id}_R \otimes \mathcal{E})(\rho)}}{I(R:A)_\rho}, \qquad 0 \le \eta_{\mathrm{MI}} \le 1. \tag{S88}$$

The product-reference coefficient relevant to ordinary mutual information is bounded by the corresponding complete conditional-mutual-information contraction coefficient [27]. For two local detector channels, sequential data processing gives

$$I(D_A : D_B) \le \eta_{\mathrm{MI}}(\mathcal{E}_A)\eta_{\mathrm{MI}}(\mathcal{E}_B) I_{\mathrm{F}}(A : B), \tag{S89}$$

and hence, for two qubits,

$$\mathcal{N}(\rho_{D_A D_B}) \le \min\left\{\frac{1}{2}, \sqrt{2\eta_A \eta_B I_{\mathrm{F}}}\right\}. \tag{S90}$$

For continuum local algebras, these coefficients must be defined with the same split or finite-resolution prescription used for I_{F}; no continuum value is assumed here.

The difficulty is that the ideal resonant Jaynes–Cummings (JC) channel need not contract mutual information uniformly. A detector initially in $|g\rangle$ maps a Fock input $|n\rangle$ to an excited-state probability

$$p_n = \sin^2(\theta\sqrt{n}). \tag{S91}$$

For the binary code $\{|0\rangle, |n\rangle\}$, the input mutual information is $\ln 2$ and the detector output mutual information is

$$I(U : D) = h_2(p_n/2) - \frac{1}{2} h_2(p_n). \tag{S92}$$

For every fixed $\theta > 0$, the sequence $\theta\sqrt{n}$ is equidistributed modulo π, so a subsequence has $p_n \to 1$. The output then retains asymptotically all of the input bit, and data processing fixes the exact worst-case coefficient at

$$\eta_{\mathrm{MI}}(\mathcal{E}_\theta) = 1. \tag{S93}$$

This is a statement about the supremum over all Fock inputs, not about the transfer efficiency for a specified squeezed state.

An average photon-number constraint alone does not restore a strict coefficient. Encode a rare bit in $|n\rangle$ with probability $q_n = E/n$ and otherwise use $|0\rangle$. The mean photon number is E, while the decoding error probability is $q_n(1 - p_n)$. Along the subsequence with $p_n \to 1$, the conditional entropy of the bit given the detector is negligible compared with its input entropy. Therefore

$$\eta_{\mathrm{MI}}^{(E)}(\mathcal{E}_\theta) = 1, \qquad E > 0. \tag{S94}$$

The same rare-code mechanism applies to any unbounded cost function. This is closely related to the average-cost obstruction for classical channel contraction established by Polyanskiy and Wu [28]. A strict uniform coefficient consequently requires a hard spectral restriction or additional channel noise; an average-cost bound by itself is insufficient.

A more informative object is the nonlinear information-dissipation curve

$$\mathfrak{F}_E(s) = \sup_{\substack{I(R:F)\le s \\ \mathrm{Tr}(\hat{n}\rho_F)\le E}} I(R:D). \tag{S95}$$

It obeys $\mathfrak{F}_E(s) \le s$. At special angles the curve is exactly known on a finite interval: if $p_{n_*} = 1$, then

$$q_* = \min\left\{\frac{E}{n_*}, \frac{1}{2}\right\}, \qquad s_* = h_2(q_*), \qquad \mathfrak{F}_E(s) = s \quad (0 \le s \le s_*). \tag{S96}$$

For $\theta = \pi/4$ one may take $n_* = 4$, and for $\theta = \pi/2$ one may take $n_* = 1$. Away from such exact transitions, the rare-code argument establishes unit limiting slope near $s = 0$, but does not determine the full curve.

A strict bound can be recovered for a noisy readout. If the detector is followed by depolarizing noise of strength p, the composite channel contains a replacement component of weight p. The quantum Doeblin bound then gives

$$\eta_{\mathrm{MI}} \le 1 - p, \qquad I(D_A : D_B) \le (1 - p_A)(1 - p_B) I_{\mathrm{F}}. \tag{S97}$$

This result follows from the quantum Doeblin decomposition and the associated conditional-mutual-information contraction theorem [27, 29]. It applies to the stated regulated depolarizing-readout model, not automatically to amplitude damping, dephasing, dark counts, or an ideal continuum Unruh–DeWitt channel.

Finally, a state-dependent estimate avoids the worst-case optimization. Applying data processing after only one local interaction yields

$$\begin{aligned} I(D_A : D_B) &\le \min\{I(D_A : B), I(A : D_B)\}, \\ \mathcal{N} &\le \min\left\{\frac{1}{2}, \sqrt{2\min\{I(D_A : B), I(A : D_B)\}}\right\}. \end{aligned} \tag{S98}$$

Unlike a uniform channel coefficient, this bound retains information about the actual field state and detector coupling. The finite-mode application is given in Supplementary Sec. S5.

References

[1] Charis Anastopoulos, Bei-Lok Hu, and Konstantina Savvidou. Quantum field theory based quantum information: Measurements and correlations. *Annals of Physics*, 450:169239, 2023.

[2] Nadia Koliopoulou, Charis Anastopoulos, and Ntina Savvidou. Quantum field theory measurements for relativistic particles. *Physical Review D*, 114:045011, 2026. doi: 10.1103/tvd2-c113.

[3] Robert Oeckl. Causal measurement in quantum field theory: Spacetime. *Physical Review D*, 113: 085002, 2026.

[4] Robert Oeckl and Adamantia Zampeli. Towards local and compositional measurements in quantum field theory. *Quantum*, 10:2232, 2026.

[5] Robin Simmons, Maria Papageorgiou, Marios Christodoulou, and Caslav Brukner. Factorization conditions and causality for local measurements in quantum field theory. *Physical Review D*, 114: 085013, 2026.

[6] Matheus H. Zambianco, Adam Teixidó-Bonfill, and Eduardo Martín-Martínez. Cosmological expansion induces interference between communication and entanglement harvesting. *Physical Review D*, 113:085023, 2026.

[7] Si-Han Li, Hui-Chen Yang, Rui-Yang Xu, and Shu-Min Wu. Does relativistic motion really freeze initially maximal entanglement? *Journal of High Energy Physics*, 2026:218, 2026.

[8] Shu-Min Wu, Si-Han Shang, Xu Han, Hui-Chen Yang, and Qianqian Liu. Topology-dependent relativistic degradation of multipartite entanglement. *The European Physical Journal C*, 86:1182, 2026.

[9] Si-Han Li, Si-Han Shang, and Shu-Min Wu. Does acceleration always degrade quantum entanglement for tetrapartite Unruh–DeWitt detectors? *Journal of High Energy Physics*, 2025:214, 2025.

[10] Shu-Min Wu, Si-Han Shang, Si-Yu Liu, Rui-Yang Xu, Qianqian Liu, and Xiao-Li Huang. Non-monotonic evolution of multipartite entanglement under the Unruh effect. *Physics Letters B*, 880: 140852, 2026.

[11] Otfried Gühne and Michael Seevinck. Separability criteria for genuine multiparticle entanglement. *New Journal of Physics*, 12(5):053002, 2010.

[12] Mengru Ma, Yinfei Li, and Jiangwei Shang. Multipartite entanglement measures: A review. *Fundamental Research*, 5(6):2489–2499, 2025.

[13] Yong-Cheng Ou and Heng Fan. Monogamy inequality in terms of negativity for three-qubit states. *Physical Review A*, 75:062308, 2007.

[14] Dong-Dong Dong, Xue-Ke Song, Liu Ye, and Dong Wang. Quantifying and probing multipartite entanglement via minimum entanglement drop. *Physical Review A*, 114:012402, 2026.

[15] Kamal Berrada and Smail Bougouffa. Impact of the Unruh effect on tripartite quantum resources in accelerated frames. *Physics Letters A*, 584:131595, 2026.

[16] Alejandro Pozas-Kerstjens and Eduardo Martín-Martínez. Entanglement harvesting from the electromagnetic vacuum with hydrogenlike atoms. *Physical Review D*, 94:064074, 2016.

[17] Donita Cohen, Y. Ben-Aryeh, and A. Mann. Transfer of correlations from squeezed states of radiation to atoms in a generalized Jaynes-Cummings model. *Physical Review A*, 49:2040–2048, 1994.

[18] César A. Agón, Horacio Casini, Umut Gürsoy, and Guim Planella Planas. Mutual Information from Modular Flow in CFTs. *Journal of High Energy Physics*, 2025(8):176, 2025.

[19] César A. Agón, Pablo Bueno, Adem Deniz Piskin, and Guido van der Velde. Mutual Information from Modular Flow in General Conformal Field Theories. *Physical Review D*, 114:L041704, 2026.

[20] Erickson Tjoa and Robert B. Mann. Harvesting correlations in Schwarzschild and collapsing shell spacetimes. *Journal of High Energy Physics*, 2020(8):155, 2020.

[21] Kensuke Gallock-Yoshimura, Erickson Tjoa, and Robert B. Mann. Harvesting entanglement with detectors freely falling into a black hole. *Physical Review D*, 104:025001, 2021.

[22] Nitesh Kumar Dubey and Sanved Kolekar. Harvesting fermionic field entanglement in Schwarzschild spacetime. *Physical Review D*, 112:025019, 2025.

[23] Abdessamie Chhieb, Chaimae Banouni, Saliha Abdessamie, and Mohamed Ouchrif. Quantum entanglement in the Dirac field quantization around charged black holes. *Physics Letters B*, 879: 140629, 2026.

[24] Satoshi Iso, Hiroshi Umetsu, and Frank Wilczek. Hawking radiation from charged black holes via gauge and gravitational anomalies. *Physical Review Letters*, 96(15):151302, 2006.

[25] J. Dimock and B. S. Kay. Scattering for massive scalar fields on Coulomb potentials and Schwarzschild metrics. *Classical and Quantum Gravity*, 3(1):71–80, 1986. doi: 10.1088/0264-9381/3/1/010.

[26] Mikhail N. Smolyakov. Asymptotic behavior of solutions and spectrum of states in the quantum scalar field theory in the Schwarzschild spacetime. *Physical Review D*, 108:105006, 2023. doi: 10.1103/PhysRevD.108.105006.

[27] Christoph Hirche, Ian George, Theshani Nuradha, and Mark M. Wilde. Conditional Contraction Coefficients and Their Applications to Quantum Networks. *Preprint*, 2026. Submitted August 2026.

[28] Yury Polyanskiy and Yihong Wu. Dissipation of Information in Channels With Input Constraints. *IEEE Transactions on Information Theory*, 62(1):35–55, 2016.

[29] Ian George, Christoph Hirche, Theshani Nuradha, and Mark M. Wilde. Quantum Doeblin Coefficients: Interpretations and Applications. *Quantum*, 10:2115, 2026.